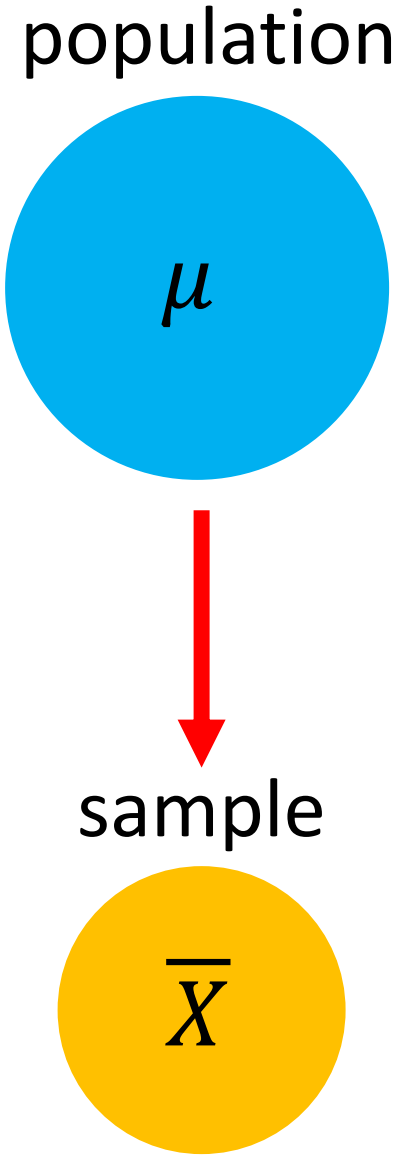
Unit 08: Hypothesis Testing

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INTRODUCTION

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Inferential Statistics

Using data to draw conclusions about a population and measuring the reliability of the conclusions.

Point Estimate

Estimate μ with $\overline{X}$

Confidence Interval

Calculate an interval of plausible values for μ

$\overline{X} \pm$ error margin = (A, B)

Hypothesis Test

Use the sample data to try to answer a question about μ

Example: Does $\mu = 8$? Yes or no?

A candy company makes bags of candies to be sold in grocery stores. The company fills bags by mass rather than by count. Bags are advertised to weigh 100 grams and contain 100 candies (on average), so **the company claims that the population mean mass among all pieces of candy = $\mu = 1$ gram**. We suspect the candy company is not fulfilling its promise of making candies that weigh 1 gram on average, and that the population mean is actually less than 1 gram.

To investigate, we take a random sample. The sample of $n = 50$ candies has a sample mean of $\overline{X}$ = 0.94 grams. Suppose the population standard deviation σ is known to be 0.19 (unrealistic).

Do we have significant evidence the population mean is actually less than 1 gram?

The candy company claims $\mu = 1$ gram.

We suspect $\mu < 1$ gram.

$H_0: \mu = 1$
$H_a: \mu < 1$

Null hypothesis = H_0 = the statement to be tested

Alternative hypothesis = H_a = the alternative to H_0

(If H_0 isn't true, what do we suspect might be true instead?)

Some courses/textbooks use H_1 instead of H_a

Since μ is an unknown parameter, we cannot know with complete certainty if H_0 is true or false. But the beauty of statistical inference is that although we may not achieve complete certainty, often we can still arrive at a conclusion and have confidence that it is probably correct.

Analogy:

Conducting a hypothesis test is like being the member of a jury at a criminal trial:

H_0 = the defendant is innocent ← Default conclusion. Often given "benefit of the doubt".

H_a = the defendant is guilty

A jury must return a verdict of guilty or not guilty. Although the jury may not have complete certainty their conclusion is correct, they are required to examine the evidence and reach the verdict.

Similarly, we have exactly two possible conclusions:

reject H_0 or **don't reject H_0**

The "evidence" we examine will be sample data. Just like the jury, we will reject H_0 iff we have convincing data (significant evidence) H_0 is false.

We have exactly two possible conclusions: reject H_0 or don't reject H_0

Instead of the phrase "don't reject" H_0, some courses/textbooks use the phrase "retain" H_0 or "fail to reject" H_0. It is not recommended however to use the phrase "accept" H_0. When we don't reject H_0, we aren't proving that H_0 is true or even saying we have a ton of confidence it is true. All we are saying is that there is not enough evidence to reject H_0.

		reality	
		H_0 is true	H_0 is false
conclusion	reject H_0	**Type 1 error**	no error
	don't reject H_0	no error	**Type 2 error**

Rejecting a true H_0 is called a **Type 1 error**.
Failing to reject a false H_0 is called a **Type 2 error**.

α = P(Type 1 error) = P(reject H_0 | H_0 is true) = significance level

β = P(Type 2 error) = P(don't reject H_0 | H_0 is false)

$1 - \beta$ = **Power** of the test = P(correctly reject H_0 | H_0 is false)

To summarize:

A researcher is interested in a population, and specifically an unknown parameter that describes the population (for example μ).

The **null hypothesis (H_0)** claims the unknown parameter equals a specific value and the researcher wants to know if this claim is true.

The **alternative hypothesis (H_a)** describes what the researcher suspects might be true instead.

After H_0 and H_a are determined, the researcher collects data.

The **test statistic** is a function of the data that helps the researcher arrive at a conclusion of "reject H_0" or "don't reject H_0".

The **rejection region** is a rule specifying for what values of the test statistic H_0 should be rejected.

Example:

Suppose your friend Billy would like to play a silly game with you over the phone. He has two baskets with colored balls inside:

Basket A
9 red
1 green

Basket B
1 red
4 green

Billy will flip a coin to select one the baskets but will not tell you the result of the coin flip. Then he will randomly select 3 balls (one at a time with replacement) from the selected basket and report the results to you over the phone. It is your job to guess what basket Billy is sampling from.

H_0: | Basket A | 9 red | 1 green |

H_a: | Basket B | 1 red | 4 green |

randomly select 3 balls (one at a time with replacement)

Rejection Region:

Let's reject H_0 iff we observe **3 green balls**.

H_0:	Basket A	H_a:	Basket B
	9 red		1 red
	1 green		4 green

randomly select 3 balls (one at a time with replacement)

Reject H_0 iff we observe 3 green balls.

α = P(Type 1 error) = P(reject H_0 | H_0 is true)

= P(GGG | Basket A) = $\left(\frac{1}{10}\right)^3 = 0.001 = \frac{1}{1000}$

H_0: | Basket A | 9 red | 1 green |

H_a: | Basket B | 1 red | 4 green |

randomly select 3 balls (one at a time with replacement)

Reject H_0 iff we observe 3 green balls.

β = P(Type 2 error) = P(don't reject H_0 | H_0 is false)

= P(anything other than GGG | Basket B) = $1 - \left(\frac{4}{5}\right)^3 = 0.488$

$1 - \beta$ = Power = P(correctly reject H_0 | H_0 is false)

$= 1 - 0.488 = 0.512$

rejection region	α P(Type 1 error)	β P(Type 2 error)	$1-\beta$ Power
3 green	0.001	0.488	0.512
2 or 3 green	0.028	0.104	0.896
at least one green	0.271	0.008	0.992

Trade offs

Recall with confidence intervals:
increasing the confidence level results in a wider interval.

With hypothesis tests, decreasing α usually results in the increase of β (and a reduction in power).

Is a completely risk-free hypothesis test ($\alpha = 0$) useful?

Recall: a 100% confidence interval was $(-\infty, \infty)$ and did not tell us anything we didn't all ready know.

With hypothesis tests, insisting on a Type 1 error probability of $\alpha = 0$ would result in $\beta = 1$. It might be nice to know a Type 1 error is impossible, but what would be the point of gathering data if you knew in advance you were not going to reject H_0 no matter what you observe?

There is no such thing as a risk-free confidence interval or hypothesis test!

AN EXAMPLE OF A HYPOTHESIS TEST FOR POPULATION MEAN

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A candy company makes bags of candies to be sold in grocery stores. The company fills bags by mass rather than by count. Bags are advertised to weigh 100 grams and contain 100 candies (on average), so **the company claims that the population mean mass among all pieces of candy = $\mu = 1$ gram**. We suspect the candy company is not fulfilling its promise of making candies that weigh 1 gram on average, and that the population mean is actually less than 1 gram. Suppose the population standard deviation σ is known to be 0.19 (unrealistic). **Do we have significant evidence the population mean is actually less than 1 gram?**

The candy company claims $\mu = 1$ gram. ← $H_0: \mu = 1$

We suspect $\mu < 1$ gram. ← $H_a: \mu < 1$

$$H_0: \mu = 1$$
$$H_a: \mu < 1$$

Plan:

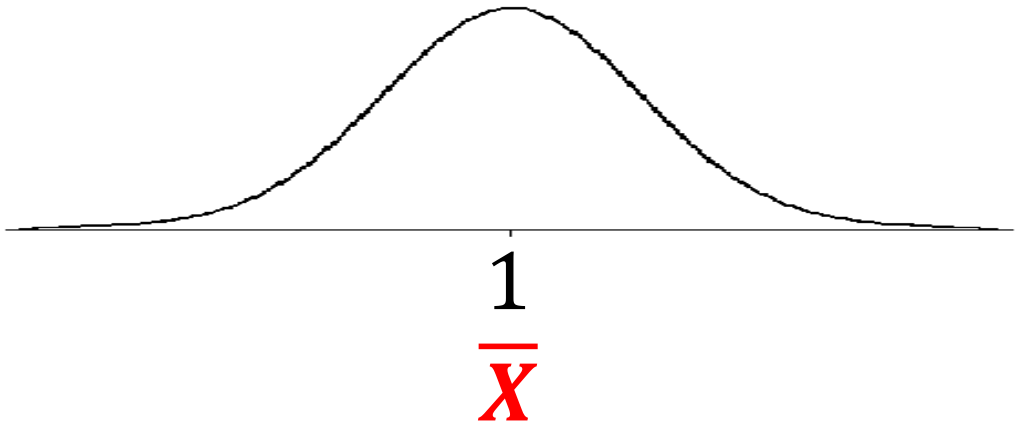
randomly select $n = 50$ candies, compute $\overline{X}$, then reject H_0 iff ___$\overline{X}$ is too low___.

Thanks to the Central Limit Theorem Part 2 and the fact that we know $\sigma = 0.19$, **if H_0 is true** then we actually know quite a bit about the $\overline{X}$ we are about to observe. It is going to come from this normal distribution: $\overline{X} \approx N\left(\mu = 1, \frac{\sigma}{\sqrt{n}} = \frac{0.19}{\sqrt{50}}\right)$

$H_0: \mu = 1$

$H_a: \mu < 1$

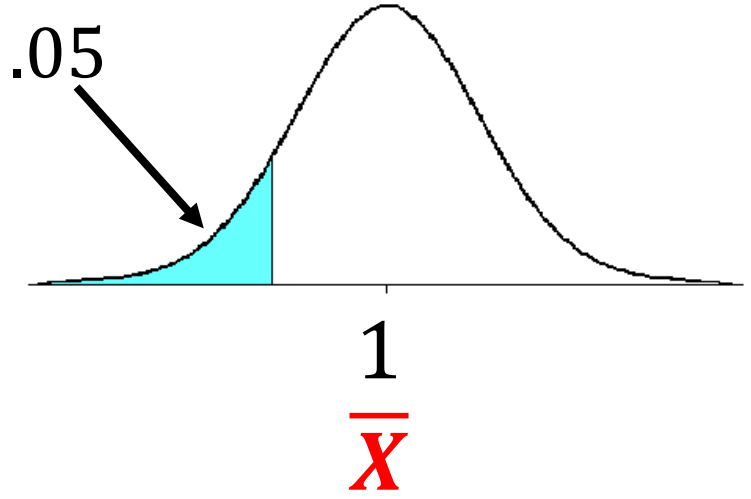
if H_0 is true, $\overline{X} \approx N\left(\mu = 1, \frac{\sigma}{\sqrt{n}} = \frac{0.19}{\sqrt{50}}\right)$



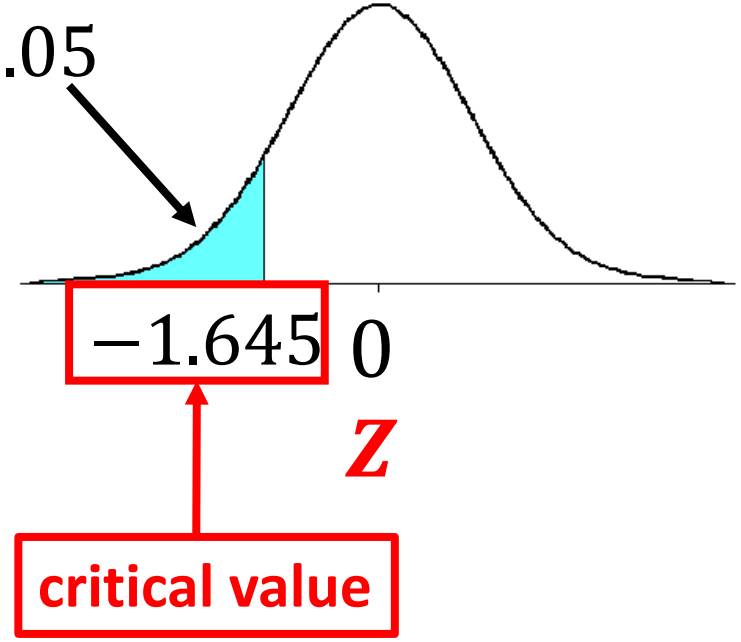
"Reject H_0 iff $\overline{X} < 1$" would be a foolish choice for a rejection region because that would result in $\alpha = .50$

What then would be a good choice for α? Let's try $\alpha = .05$

Identify a rejection region in the left tail with an area of $\alpha = .05$



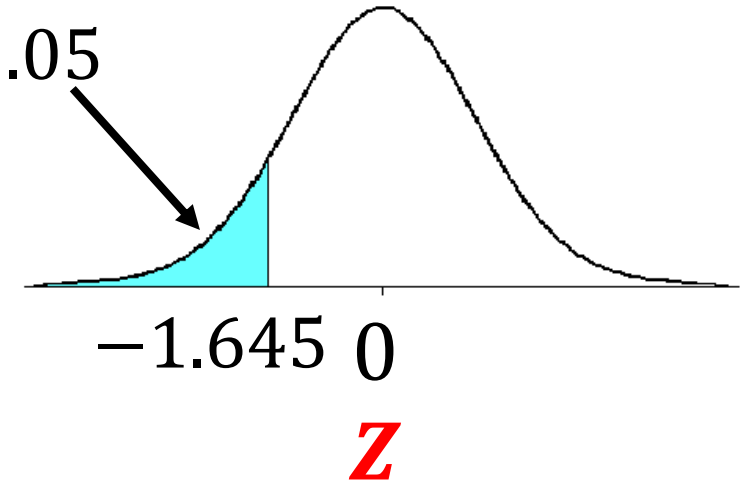
For convenience, let's find a z-score for the observed sample mean and work directly with the Z curve.



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$H_0: \mu = 1$

$H_a: \mu < 1$



Plan: randomly select $n = 50$ candies, compute $\overline{X}$, find its z-score $Z = \frac{\overline{X}-\mu}{\sigma/\sqrt{n}} = \frac{\overline{X}-1}{0.19/\sqrt{50}}$, then reject H_0 iff $Z < -1.645$

Given: suppose $\overline{X} = 0.94$ grams. $Z = \frac{0.94-1}{0.19/\sqrt{50}} \approx -2.23$

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Conclusion: since the test statistic is inside the rejection region, we reject H_0. We have significant evidence that the population mean mass among the candies is less than 1 gram.

Review:

Null and Alternative hypothesis:

$H_0: \mu = 1$

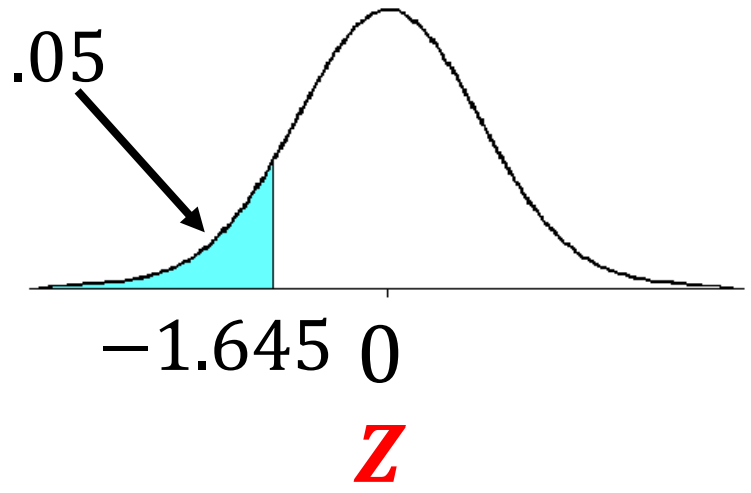
$H_a: \mu < 1$

Test statistic:

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{0.94 - 1}{0.19/\sqrt{50}} \approx -2.23$$

Rejection Region ($\alpha = .05$)

RR: $Z < -1.645$



Conclusion: reject H_0

Null and alternative hypothesis:

$H_0: \mu = 1$

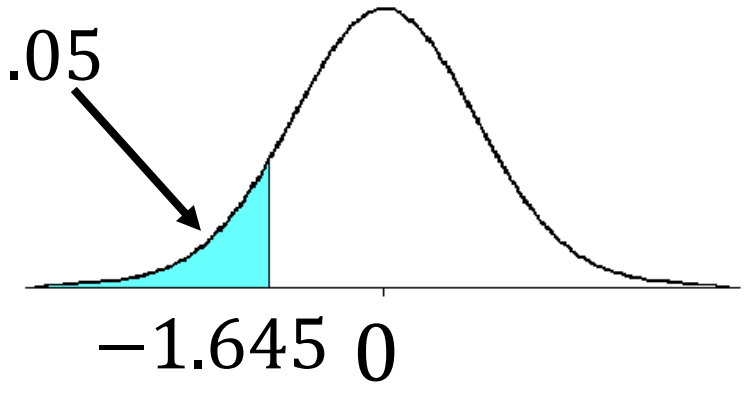
$H_a: \mu < 1$

Test statistic:

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{0.94 - 1}{0.19/\sqrt{50}} \approx -2.23$$

Rejection Region ($\alpha = .05$)

RR: $Z < -1.645$



Conclusion: reject H_0

Hypothetical Question:
what if μ really does equal 1?

Then we committed a Type 1 error.

Hypothetical Question:
what if μ actually equals 0.97?

Then we did not commit an error.

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ONE-SAMPLE Z TEST FOR POPULATION MEAN WHEN SIGMA IS KNOWN

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In the previous topic we wanted to know if the mean mass in a population of candies was $\mu = 1$ gram and suspected that maybe μ was less than 1 gram. We collected data from a random sample and concluded we did have significant evidence that $\mu < 1$.

The hypothesis test we did was a

one-sample Z test for μ when σ is known

and we went through several steps.

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one-sample Z test for μ when σ is known

1

Determine the null and alternative hypothesis.

$H_0: \mu = k$

$H_a: \mu < k$ or $H_a: \mu > k$ or $H_a: \mu \neq k$

left-tailed | right-tailed | two-tailed

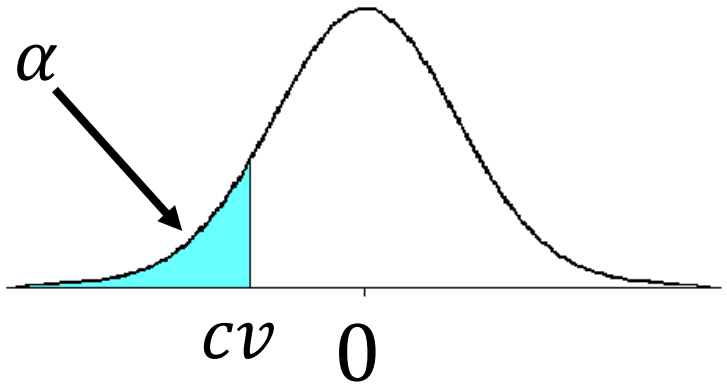
2

Choose the significance level α = P(Type 1 error).

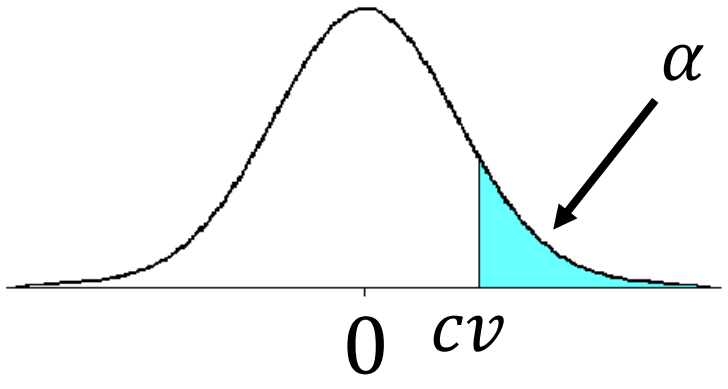
one-sample Z test for μ when σ is known

3

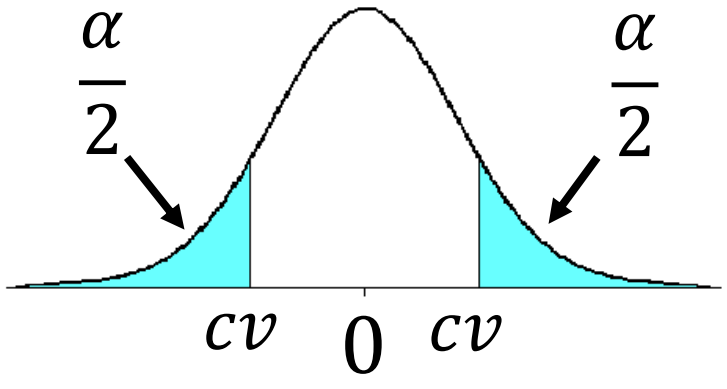
Draw the rejection region. area = α. location $\leftrightarrow$ H_a



left-tailed



right-tailed



two-tailed

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one-sample Z test for μ when σ is known

4

Collect data and calculate the test statistic.

$$Z = \frac{\overline{X} - \mu}{\sigma / \sqrt{n}}$$

5

Conclusion. If the test statistic lands inside the rejection region, then reject H_0 and say "we have significant evidence H_0 is false". Otherwise, don't reject H_0.

one-sample Z test for μ when σ is known

6

Draw a picture to find and interpret the p-value.
(next topic!)

P-VALUES

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Recall the hypothesis test about the mass of the candies:

Null and alternative hypothesis:

$H_0: \mu = 1$

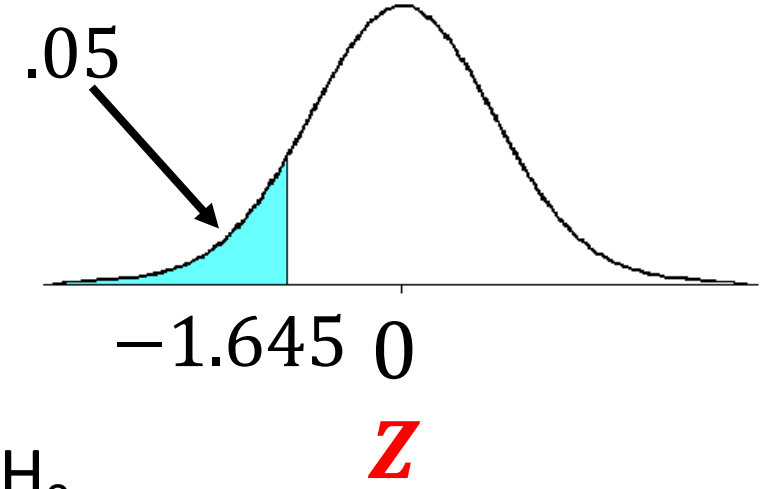
$H_a: \mu < 1$

Test statistic:

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{0.94 - 1}{0.19/\sqrt{50}} \approx -2.23$$

Rejection Region ($\alpha = .05$)

RR: $Z < -1.645$



Conclusion: reject H_0

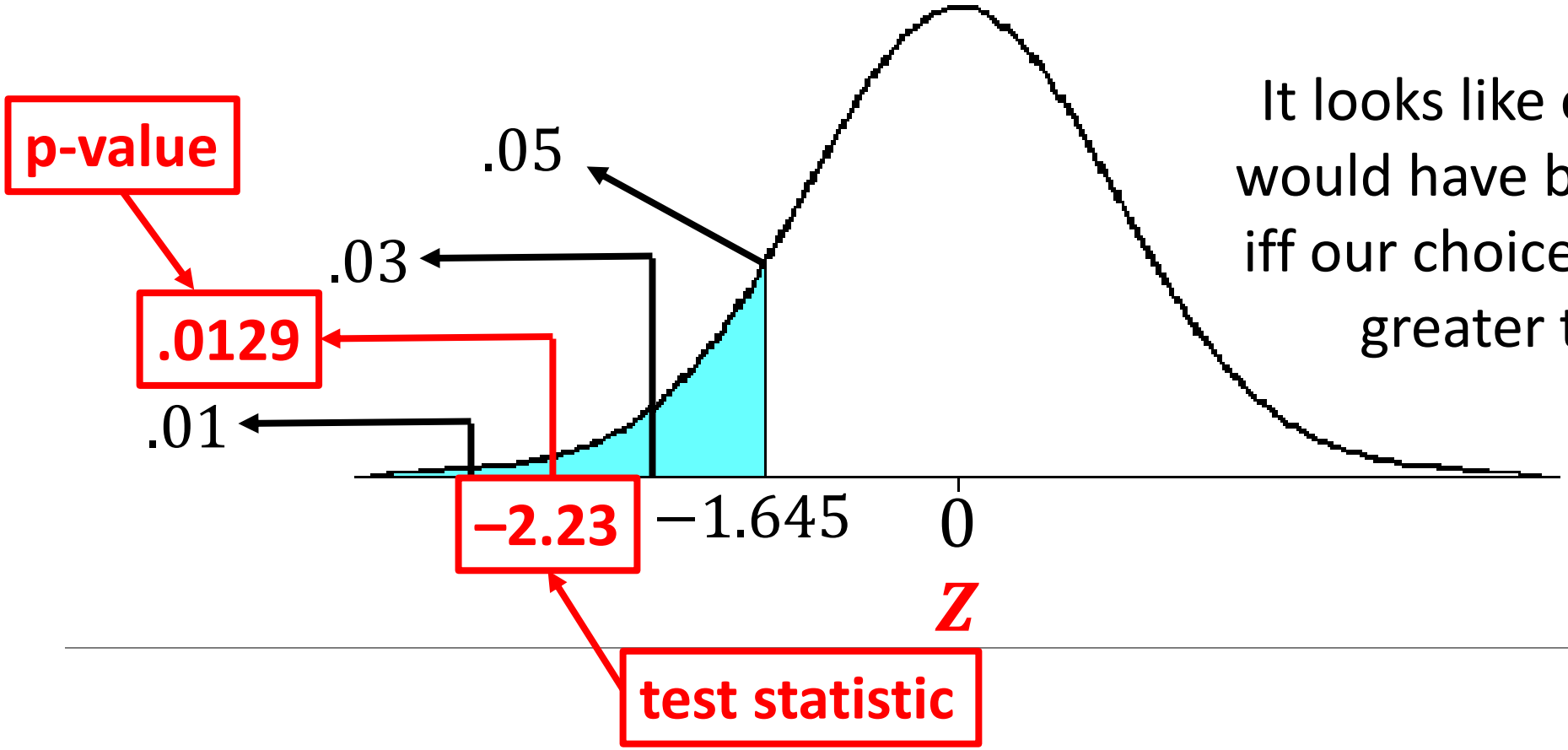
Let's take a close look at the rejection region and ask some hypothetical questions.

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Test statistic: $Z = \frac{\overline{X}-\mu}{\sigma/\sqrt{n}} = \frac{0.94-1}{0.19/\sqrt{50}} \approx -2.23$

What if α had been .03 instead of .05? We still would have rejected H_0

What if α had been .01 instead of .05? We would not have rejected H_0!

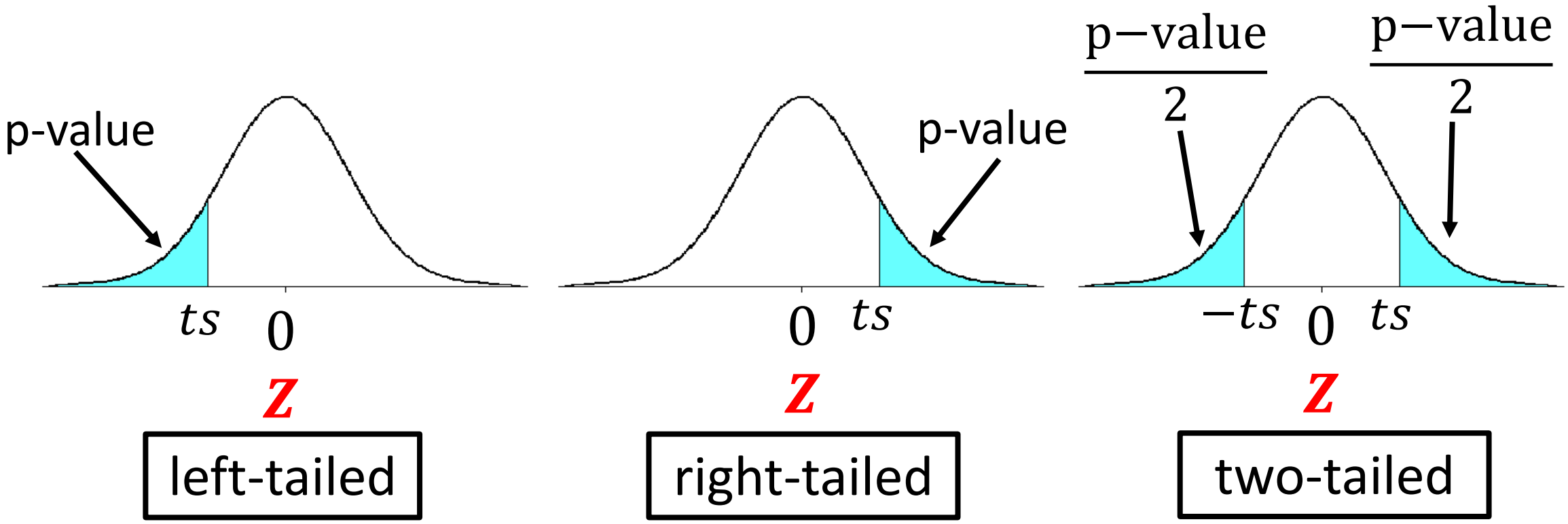


It looks like our conclusion would have been "reject H_0" iff our choice of α had been greater than .0129

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p-value

The p-value is the area further into the tail(s) from the test statistic:



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p-value

The p-value measures how significant your data is: the closer it is to zero, the more evidence you have against H_0.

The sample mean in the story was $\overline{X} = 0.94$ grams. Imagine instead it had been a much smaller value like 0.85 grams.

$H_0: \mu = 1$
$H_a: \mu < 1$

Original test statistic:

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{0.94 - 1}{0.19/\sqrt{50}} \approx -2.23$$

Original p-value:

.0129

Imagined test statistic:

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{0.85 - 1}{0.19/\sqrt{50}} \approx -5.58$$

Imagined p-value:

$$\approx \frac{1}{83{,}000{,}000}$$

p-value

Casual definition: the p-value is the probability of observing data as extreme (or more extreme) as what you observed in your random sample **if H_0 is true**.

$H_0: \mu = 1$
$H_a: \mu < 1$

test statistic:

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{0.94 - 1}{0.19/\sqrt{50}} \approx -2.23$$

p-value:
.0129

So if H_0 is true and μ really does equal 1 gram, then only about 1.29% of random samples of size $n = 50$ will have a sample mean of 0.94 grams or less (or equivalently, a test statistic of $Z = -2.23$ or less).

p-value

Remember how a hypothesis test is like a criminal trial? In real life, a jury is doing a hypothesis test even though they don't use the term.

H_0: defendant is innocent

H_a: defendant is guilty

Sometimes the criminal trial even has a p-value! The prosecuting attorney could ask an expert witness to testify about the probability the defendant's DNA and the killer's DNA would be a match just by chance, assuming the defendant is innocent.

The extremely small match probability can be considered a p-value.

p-value

Every hypothesis test must follow this rule:

H_0 is rejected iff p-value $< \alpha$

H_0 is not rejected iff p-value $> \alpha$

Some courses and texts don't bother with rejection regions at all and simply refer to this rule whenever a conclusion is desired.

p-value

Alan and Ben each do a hypothesis test independently of each other.

Coincidentally, they both chose a significance level of $\alpha = .08$

Alan's p-value = .04

Ben's p-value = $\frac{1}{1{,}000{,}000}$

Both Alan and Ben rejected their null hypothesis, but Ben is more confident in his rejection.

Null and alternative hypothesis:

$H_0: \mu = 50$

$H_a: \mu \neq 50$

two-tailed test

data:

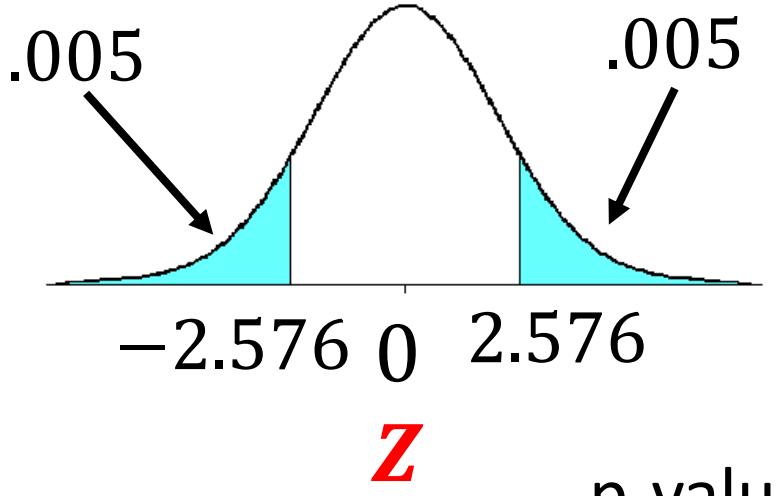
$n = 64, \overline{X} = 53, \sigma = 16$ (known)

Test statistic:

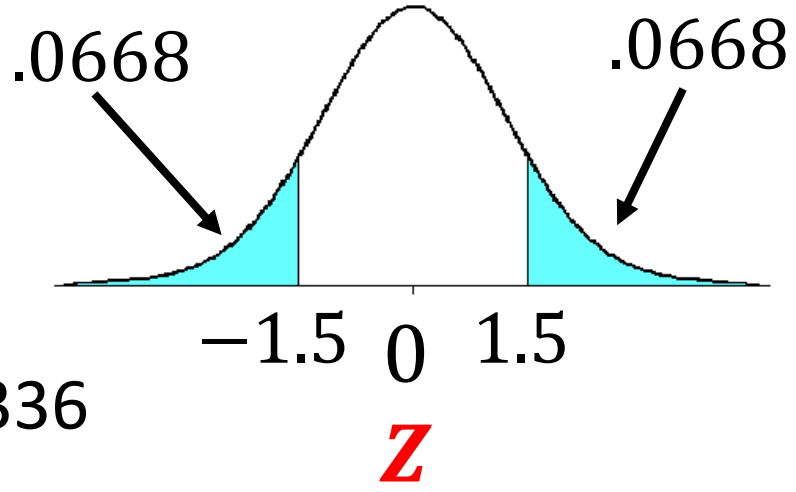
$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{53 - 50}{16/\sqrt{64}} = 1.5$$

Rejection Region ($\alpha = .01$)

RR: $Z < -2.576$ or $Z > 2.576$



Conclusion: don't reject H_0



p-value = 2(.0668) = .1336

p-value $> \alpha$

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μ = the population mean weight among all Michigan Tech students. We want to know if $\mu = 140$, and suspect it may be greater than 140. Conduct a hypothesis test with $\alpha = .0418$

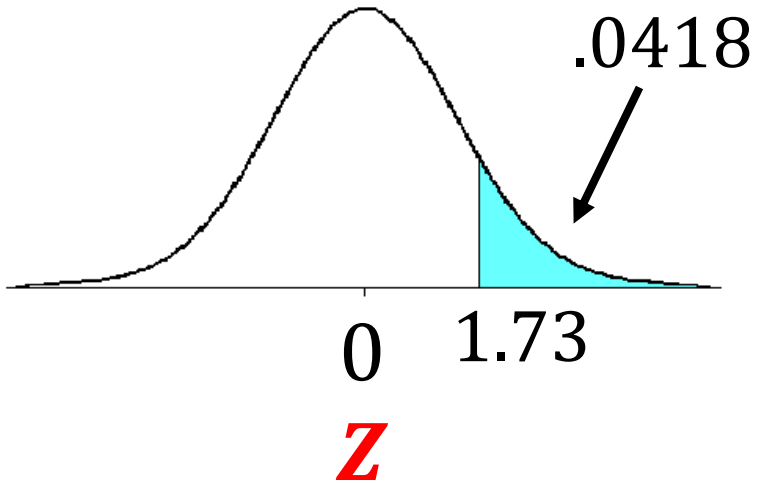
H_0: $\mu = 140$

H_a: $\mu > 140$

right-tailed test

Rejection Region ($\alpha = .0418$)

RR: $Z > 1.73$



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μ = the population mean weight among all Michigan Tech students. We want to know if $\mu = 140$, and suspect it may be greater than 140. Conduct a hypothesis test with $\alpha = .0418$

H_0: $\mu = 140$
H_a: $\mu > 140$

Rejection Region ($\alpha = .0418$)
RR: $Z > 1.73$

data:
$n = 30, \overline{X} = 150, \sigma = 25$ (known)

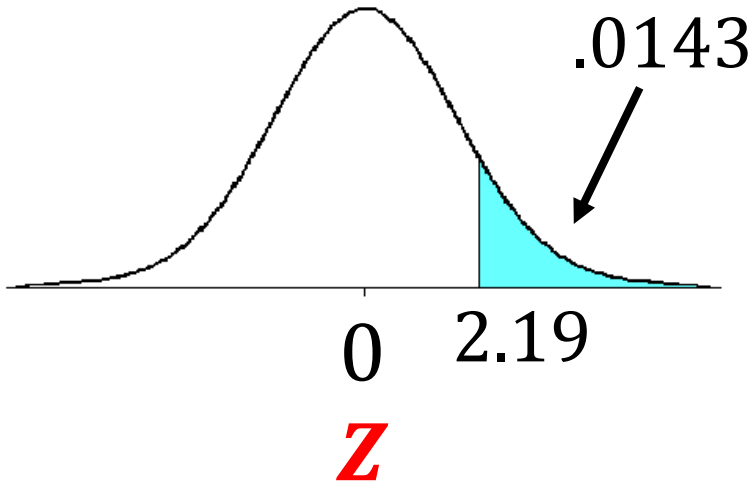
Test statistic:
$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{150 - 140}{25/\sqrt{30}} \approx 2.19$$

Conclusion: reject H_0

Conclusion: reject H_0

p-value = .0143

p-value $< \alpha$

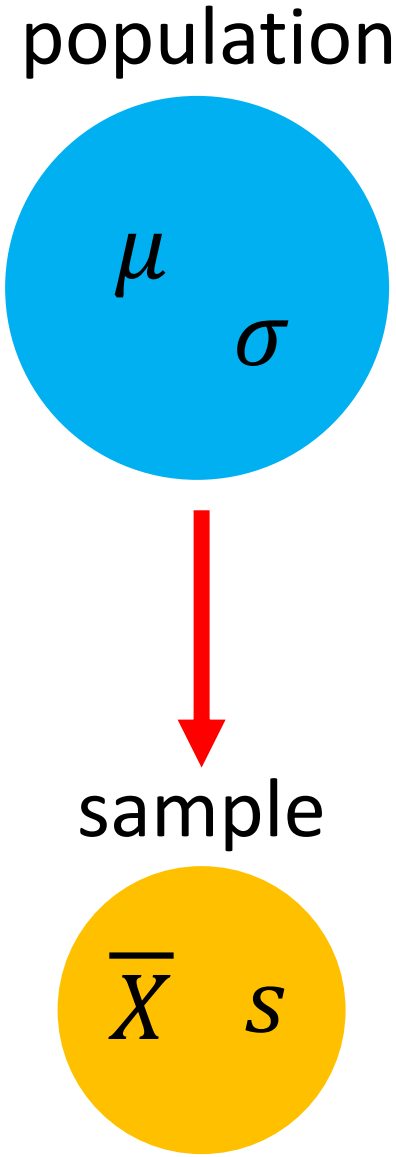


p-value interpretation:

If H_0 is true and μ really does equal 140 pounds, then only about 1.43% of random samples of size $n = 30$ will have a sample mean of 150 pounds or more (or equivalently, a test statistic of $Z = 2.19$ or more).

ONE SAMPLE T TEST FOR POPULATION MEAN WHEN SIGMA IS UNKNOWN

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If μ is unknown, then σ would most unlikely be unknown as well. Now we will consider this more realistic situation.

In addition to estimating μ with $\overline{X}$, we will estimate σ with s.

A $(1-\alpha)100\%$ confidence interval for μ when **σ is known** (population is normally distributed or $n \geq 30$)

$$\overline{X} \pm Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

A $(1-\alpha)100\%$ confidence interval for μ when **σ is unknown** (population is normally distributed or $n \geq 30$)

$$\overline{X} \pm t_{n-1,\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$$

Important!

When doing a one-sample confidence interval or hypothesis test for μ:

Use the Z curve if **σ is known**

Use the T curve if **σ is unknown**

Our policy in this course:

When doing a one-sample confidence interval or hypothesis test for μ:

Use the Z curve if **σ is known**

Use the T curve if **σ is unknown**

Warning

Many courses/textbooks use this slightly different policy.

(Technically, this is incorrect.)

	$n \geq 30$	$n < 30$
σ known	**Z**	**Z**
σ unknown	**Z (close enough)**	**T**

A **one sample T test for μ when σ is unknown** is extremely similar to a one sample Z test for μ when σ is known , but the test statistic will be

$$T = \frac{\overline{X} - \mu}{s/\sqrt{n}}$$

Also, the rejection region and the picture we draw for the p-value will be on a T curve with $n - 1$ degrees of freedom.

Investigating the p-value will take a little more work.

μ = population mean cholesterol level (mg/dL) among 70-year-old American men. We want to know if μ = 42.5 mg/dL and we suspect μ may be greater than 42.5 mg/dL. Conduct a hypothesis test with $\alpha = .05$

$H_0: \mu = 42.5$

$H_a: \mu > 42.5$

$n = 19$

$\overline{X} = 44.2$ mg/dL

$s = 3.8$ (σ unknown)

→ Small sample size, therefore we must assume the population of cholesterol levels are normally distributed and we should examine a QQ plot. (CLT Part 1)

$H_0: \mu = 42.5$

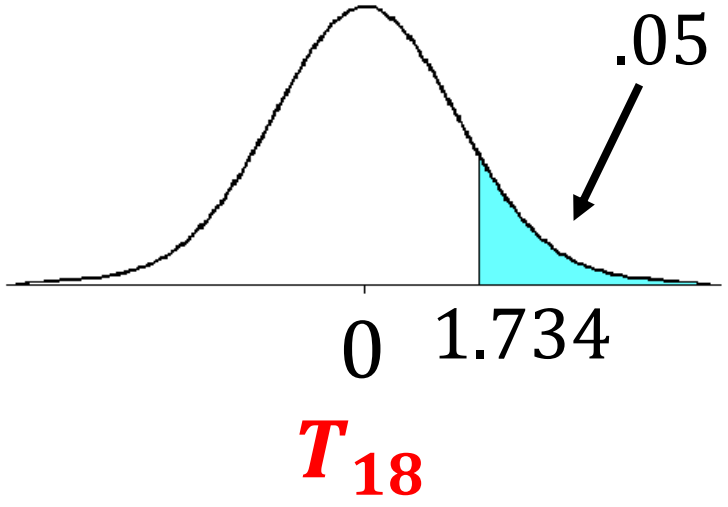
$H_a: \mu > 42.5$

Test statistic:

$$T = \frac{\overline{X} - \mu}{s/\sqrt{n}} = \frac{44.2 - 42.5}{3.8/\sqrt{19}} \approx 1.95$$

Rejection Region ($\alpha = .05$)

RR: $T > 1.734$



Conclusion: reject H_0

Since we rejected H_0, we know in advance

p-value $< \alpha = .05$

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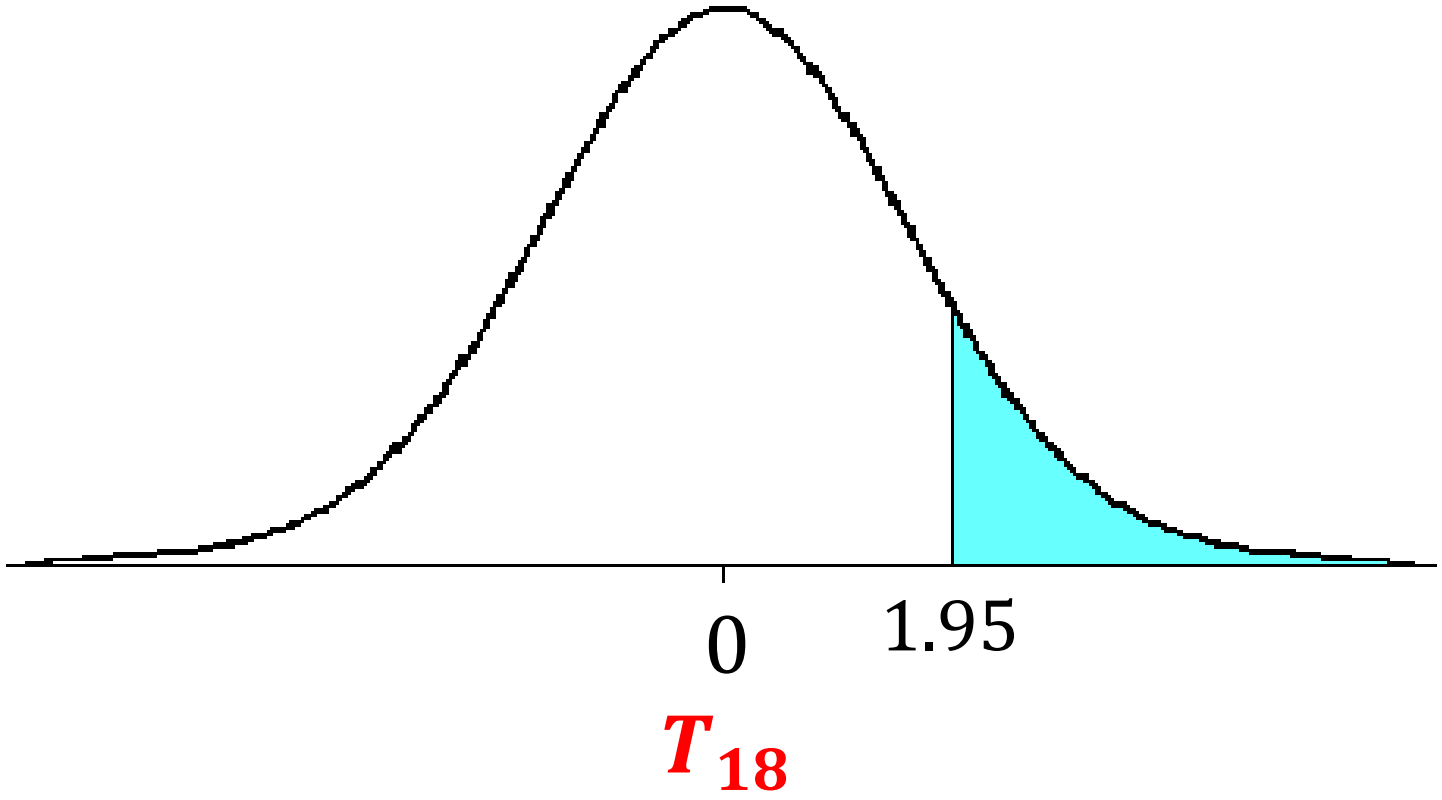
$H_0: \mu = 42.5$

$H_a: \mu > 42.5$

Test statistic:

$T \approx 1.95$

Since we rejected H_0, we know in advance p-value $< \alpha = .05$



Look for two values in row 18. One value should be slightly less than 1.95, the other value should be slightly greater than 1.95.

T-Table



	0.20	0.10	0.05	0.025	0.01	0.005	0.001	0.0005	0.0001
1	1.376	3.078	6.314	12.706	31.821	63.657	318.309	636.619	3183.099
2	1.061	1.886	2.920	4.303	6.965	9.925	22.327	31.599	70.700
3	0.978	1.638	2.353	3.182	4.541	5.841	10.215	12.924	22.204
4	0.941	1.533	2.132	2.776	3.747	4.604	7.173	8.610	13.034
5	0.920	1.476	2.015	2.571	3.365	4.032	5.893	6.869	9.678
6	0.906	1.440	1.943	2.447	3.143	3.707	5.208	5.959	8.025
7	0.896	1.415	1.895	2.365	2.998	3.499	4.785	5.408	7.063
8	0.889	1.397	1.860	2.306	2.896	3.355	4.501	5.041	6.442
9	0.883	1.383	1.833	2.262	2.821	3.250	4.297	4.781	6.010
10	0.879	1.372	1.812	2.228	2.764	3.169	4.144	4.587	5.694
11	0.876	1.363	1.796	2.201	2.718	3.106	4.025	4.437	5.453
12	0.873	1.356	1.782	2.179	2.681	3.055	3.930	4.318	5.263
13	0.870	1.350	1.771	2.160	2.650	3.012	3.852	4.221	5.111
14	0.868	1.345	1.761	2.145	2.624	2.977	3.787	4.140	4.985
15	0.866	1.341	1.753	2.131	2.602	2.947	3.733	4.073	4.880
16	0.865	1.337	1.746	2.120	2.583	2.921	3.686	4.015	4.791
17	0.863	1.333	1.740	2.110	2.567	2.898	3.646	3.965	4.714
18	0.862	1.330	1.734	2.101	2.552	2.878	3.610	3.922	4.648
19	0.861	1.328	1.729	2.093	2.539	2.861	3.579	3.883	4.590
20	0.860	1.325	1.725	2.086	2.528	2.845	3.552	3.850	4.539

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$H_0: \mu = 42.5$

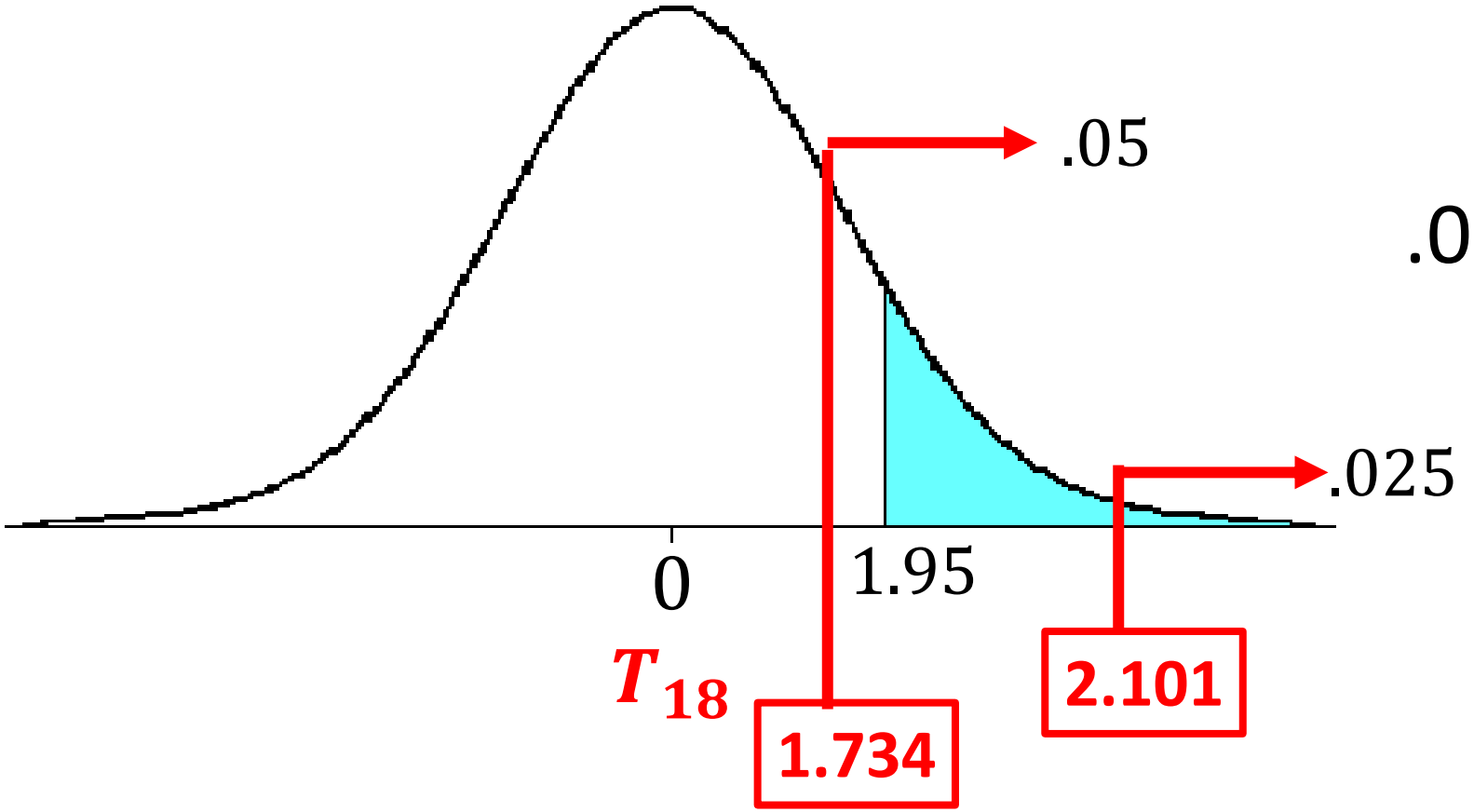
$H_a: \mu > 42.5$

Test statistic:

$T \approx 1.95$

Since we rejected H_0, we know in advance

p-value $< \alpha = .05$



.025 < p-value < .05

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P-VALUE BOUNDS ON T CURVES

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think beyond the possible℠

$H_0: \mu = 42.5$

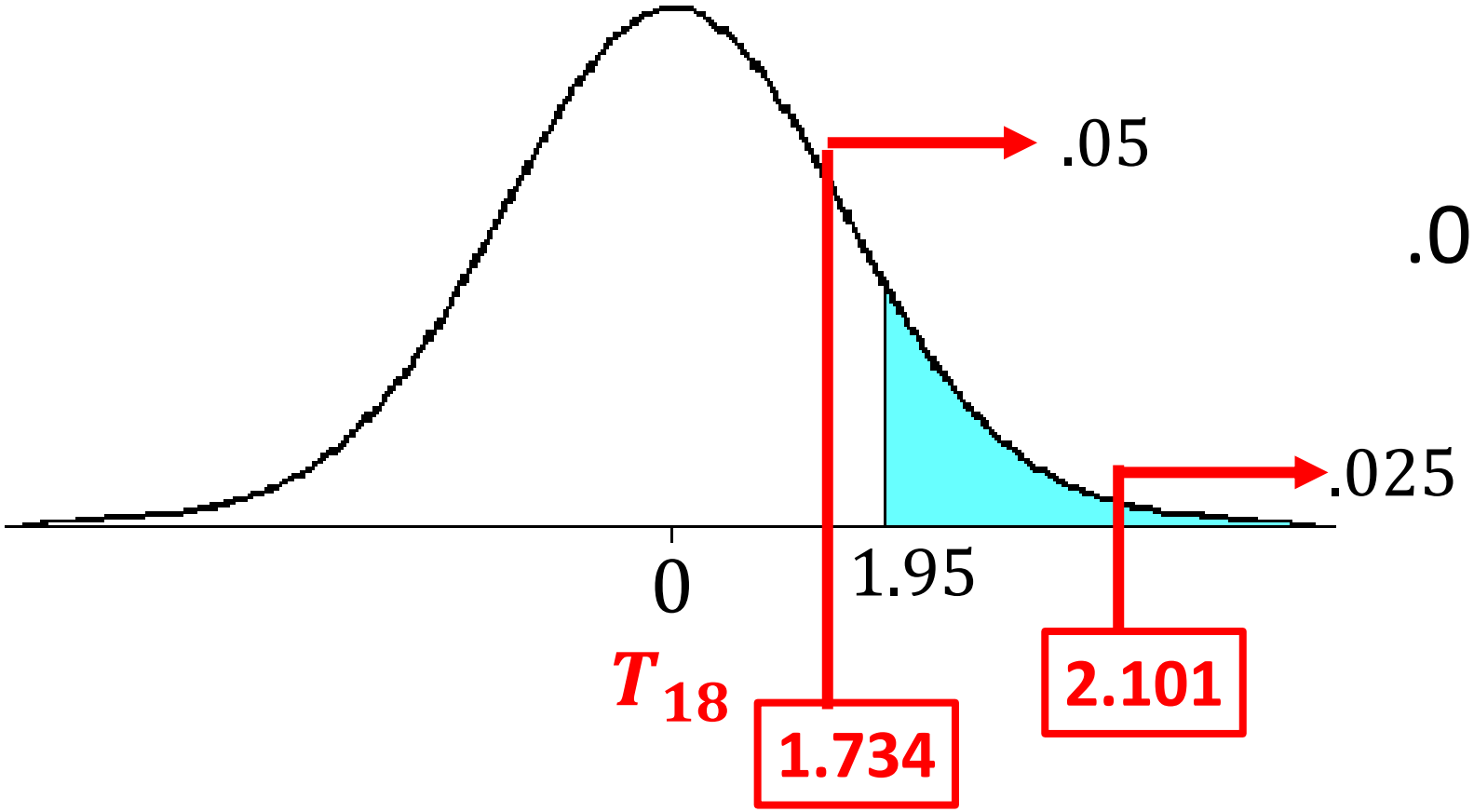
$H_a: \mu > 42.5$

Test statistic:

$T \approx 1.95$

Since we rejected H_0, we know in advance

p-value $< \alpha = .05$



.025 < p-value < .05

CASE WESTERN RESERVE UNIVERSITY

$H_0: \mu = 10.3$

$H_a: \mu < 10.3$

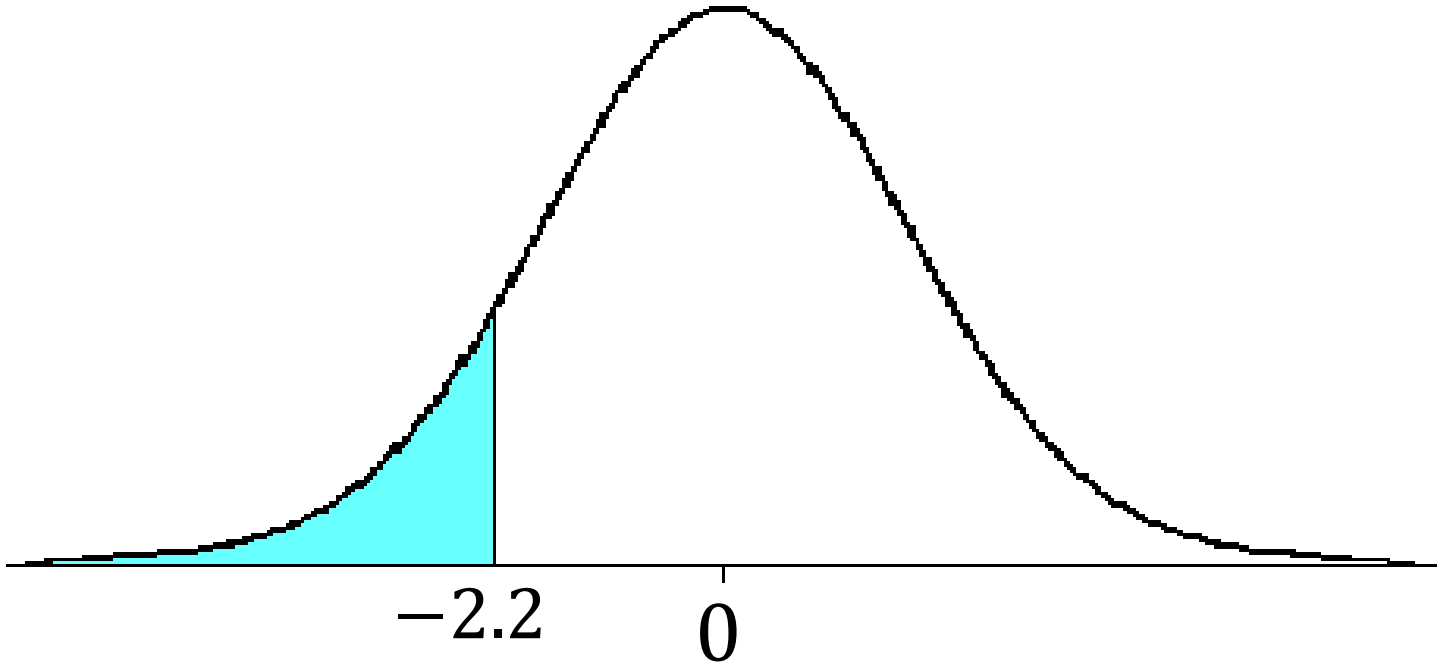
left-tailed test

Test statistic:

$$T = \frac{\overline{X} - \mu}{s/\sqrt{n}} = -2.2$$

Sample size:

$n = 15$



CASE WESTERN RESERVE UNIVERSITY

Look for two values in row 14. One value should be slightly less than 2.2, the other value should be slightly greater than 2.2

T-Table



	0.20	0.10	0.05	0.025	0.01	0.005	0.001	0.0005	0.0001
1	1.376	3.078	6.314	12.706	31.821	63.657	318.309	636.619	3183.099
2	1.061	1.886	2.920	4.303	6.965	9.925	22.327	31.599	70.700
3	0.978	1.638	2.353	3.182	4.541	5.841	10.215	12.924	22.204
4	0.941	1.533	2.132	2.776	3.747	4.604	7.173	8.610	13.034
5	0.920	1.476	2.015	2.571	3.365	4.032	5.893	6.869	9.678
6	0.906	1.440	1.943	2.447	3.143	3.707	5.208	5.959	8.025
7	0.896	1.415	1.895	2.365	2.998	3.499	4.785	5.408	7.063
8	0.889	1.397	1.860	2.306	2.896	3.355	4.501	5.041	6.442
9	0.883	1.383	1.833	2.262	2.821	3.250	4.297	4.781	6.010
10	0.879	1.372	1.812	2.228	2.764	3.169	4.144	4.587	5.694
11	0.876	1.363	1.796	2.201	2.718	3.106	4.025	4.437	5.453
12	0.873	1.356	1.782	2.179	2.681	3.055	3.930	4.318	5.263
13	0.870	1.350	1.771	2.160	2.650	3.012	3.852	4.221	5.111
14	[illegible]	[illegible]	[illegible]	2.145	2.624	2.977	3.787	4.140	4.985
15	0.866	1.341	1.753	2.131	2.602	2.947	3.733	4.073	4.880
16	0.865	1.337	1.746	2.120	2.583	2.921	3.686	4.015	4.791
17	0.863	1.333	1.740	2.110	2.567	2.898	3.646	3.965	4.714
18	0.862	1.330	1.734	2.101	2.552	2.878	3.610	3.922	4.648
19	0.861	1.328	1.729	2.093	2.539	2.861	3.579	3.883	4.590
20	0.860	1.325	1.725	2.086	2.528	2.845	3.552	3.850	4.539

CASE WESTERN RESERVE UNIVERSITY

$H_0: \mu = 10.3$

$H_a: \mu < 10.3$

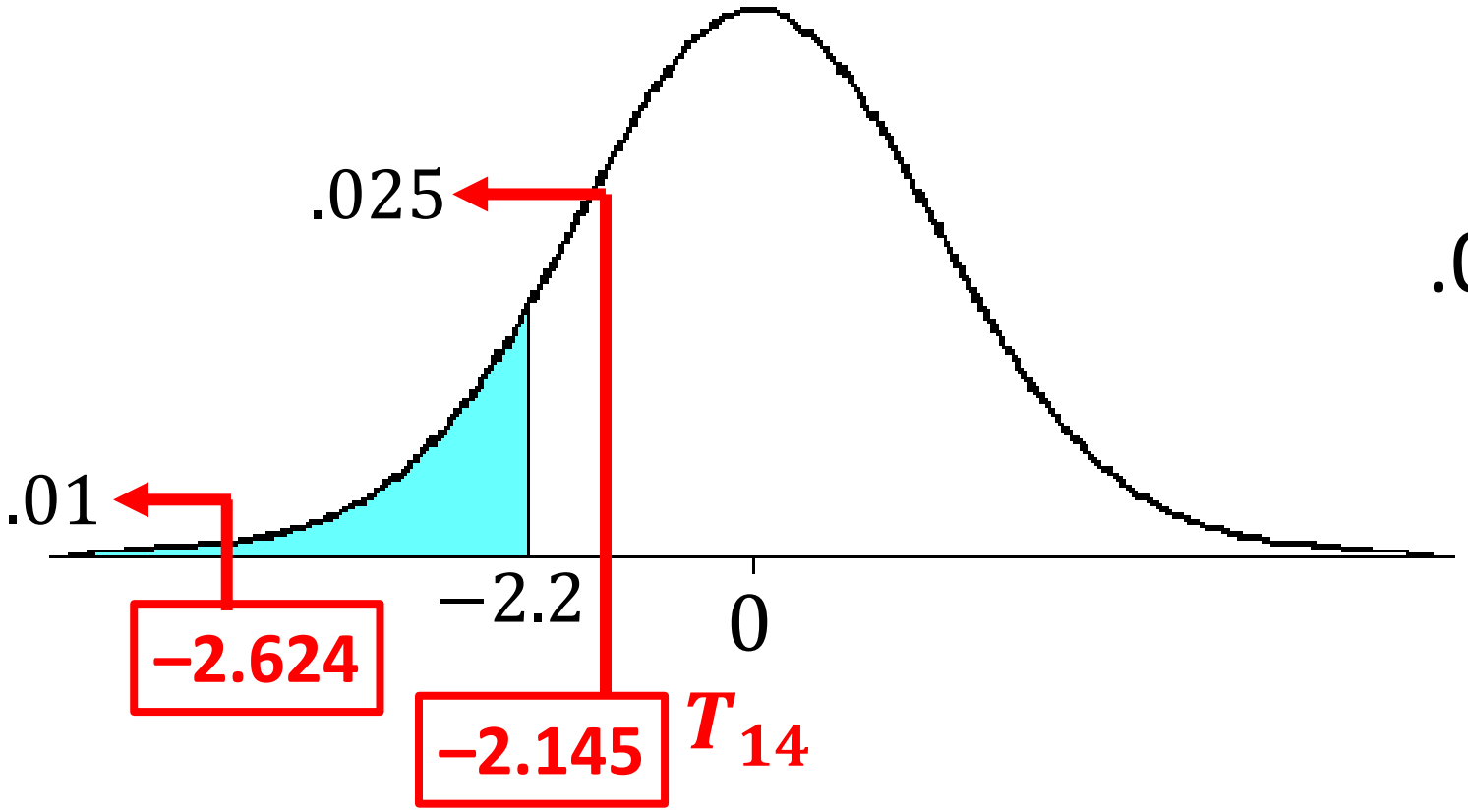
left-tailed test

Test statistic:

$$T = \frac{\overline{X} - \mu}{s/\sqrt{n}} = -2.2$$

Sample size:

$n = 15$



.01 < p-value < .025

CASE WESTERN RESERVE UNIVERSITY

$H_0: \mu = 88$

$H_a: \mu > 88$

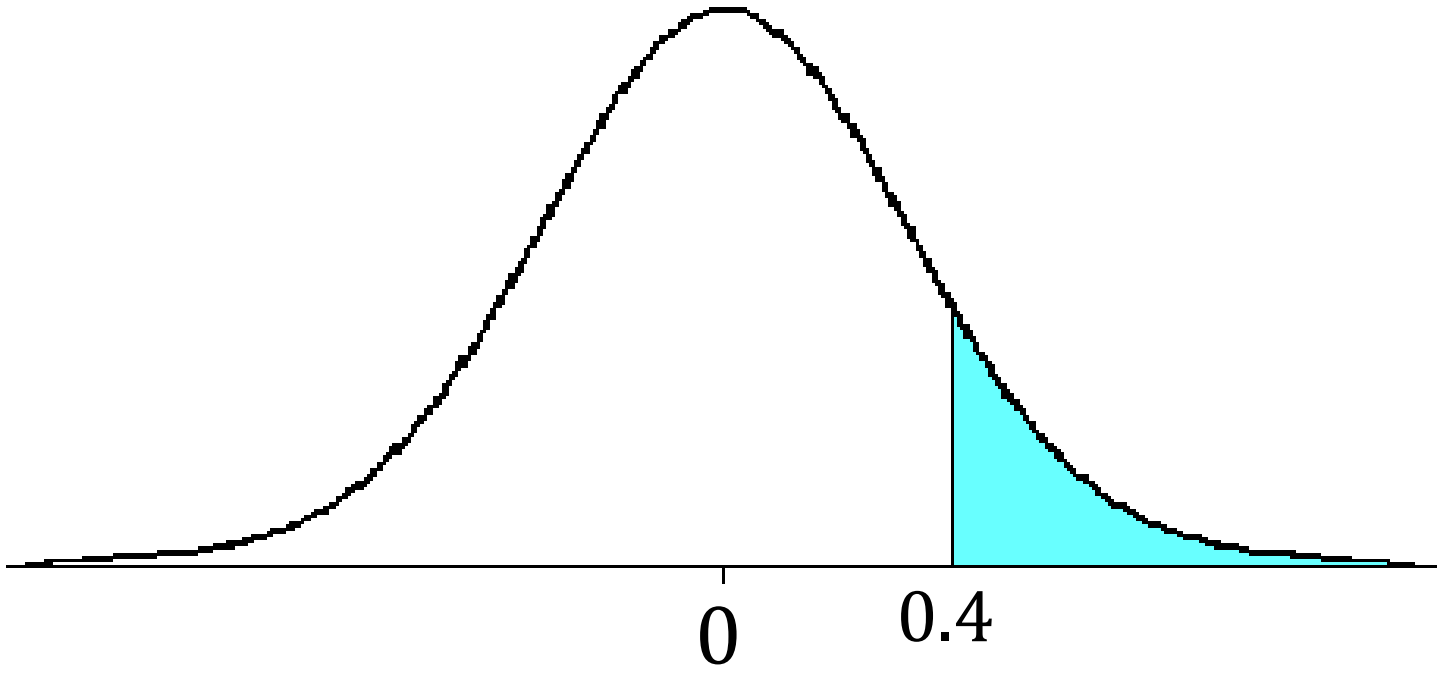
right-tailed test

Test statistic:

$$T = \frac{\overline{X} - \mu}{s/\sqrt{n}} = 0.4$$

Sample size:

$n = 11$



T_{10}

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All the t values in row 10 are greater than 0.4 !!

T-Table



	0.20	0.10	0.05	0.025	0.01	0.005	0.001	0.0005	0.0001
1	1.376	3.078	6.314	12.706	31.821	63.657	318.309	636.619	3183.099
2	1.061	1.886	2.920	4.303	6.965	9.925	22.327	31.599	70.700
3	0.978	1.638	2.353	3.182	4.541	5.841	10.215	12.924	22.204
4	0.941	1.533	2.132	2.776	3.747	4.604	7.173	8.610	13.034
5	0.920	1.476	2.015	2.571	3.365	4.032	5.893	6.869	9.678
6	0.906	1.440	1.943	2.447	3.143	3.707	5.208	5.959	8.025
7	0.896	1.415	1.895	2.365	2.998	3.499	4.785	5.408	7.063
8	0.889	1.397	1.860	2.306	2.896	3.355	4.501	5.041	6.442
9	0.883	1.383	1.833	2.262	2.821	3.250	4.297	4.781	6.010
10	0.879	1.372	1.812	2.228	2.764	3.169	4.144	4.587	5.694
11	0.876	1.363	1.796	2.201	2.718	3.106	4.025	4.437	5.453
12	0.873	1.356	1.782	2.179	2.681	3.055	3.930	4.318	5.263
13	0.870	1.350	1.771	2.160	2.650	3.012	3.852	4.221	5.111
14	0.868	1.345	1.761	2.145	2.624	2.977	3.787	4.140	4.985
15	0.866	1.341	1.753	2.131	2.602	2.947	3.733	4.073	4.880
16	0.865	1.337	1.746	2.120	2.583	2.921	3.686	4.015	4.791
17	0.863	1.333	1.740	2.110	2.567	2.898	3.646	3.965	4.714
18	0.862	1.330	1.734	2.101	2.552	2.878	3.610	3.922	4.648
19	0.861	1.328	1.729	2.093	2.539	2.861	3.579	3.883	4.590
20	0.860	1.325	1.725	2.086	2.528	2.845	3.552	3.850	4.539

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$H_0: \mu = 88$

$H_a: \mu > 88$

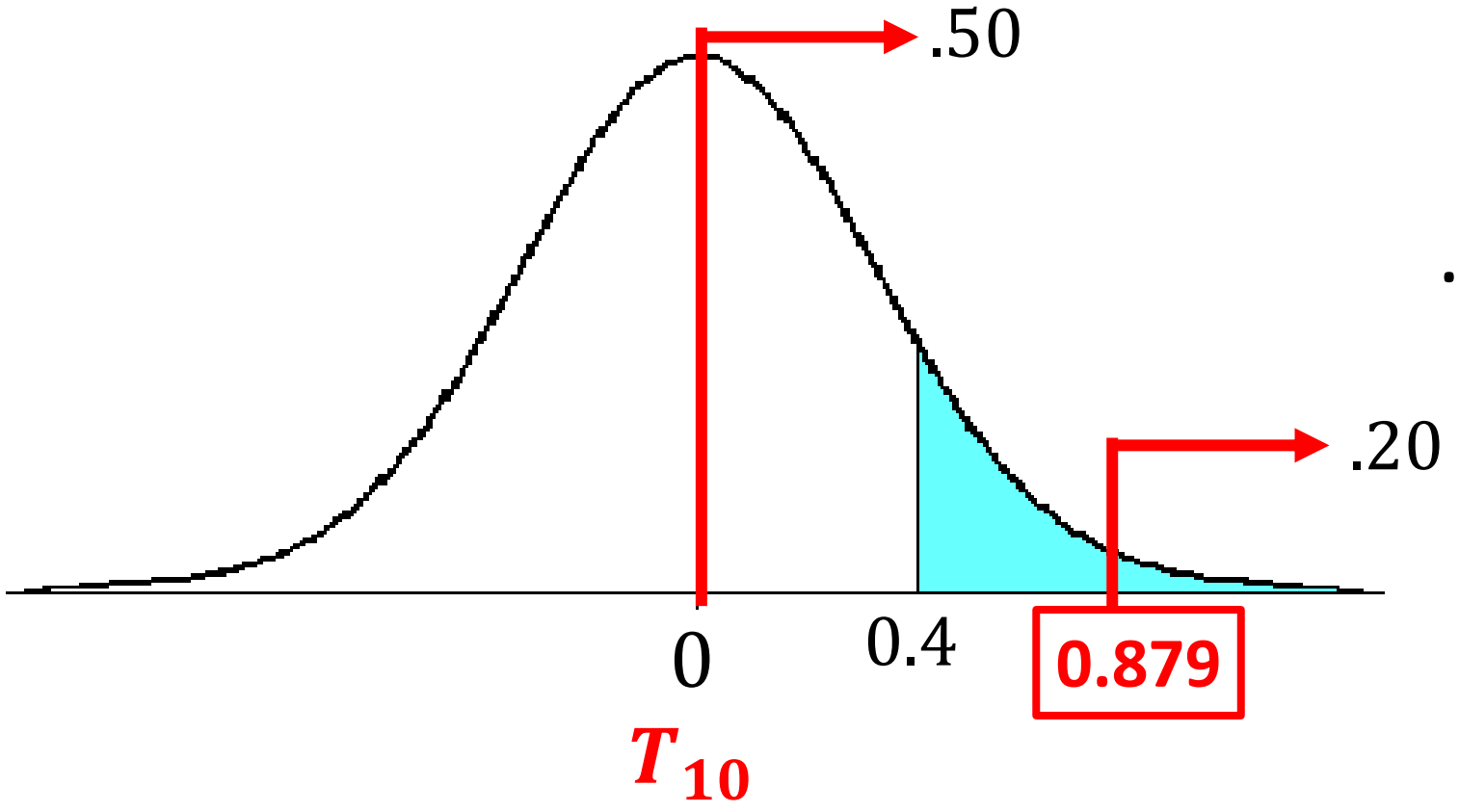
right-tailed test

Test statistic:

$$T = \frac{\overline{X} - \mu}{s/\sqrt{n}} = 0.4$$

Sample size:

$n = 11$



.20 < p-value < .50

CASE WESTERN RESERVE UNIVERSITY

$H_0: \mu = 3.72$

$H_a: \mu \neq 3.72$

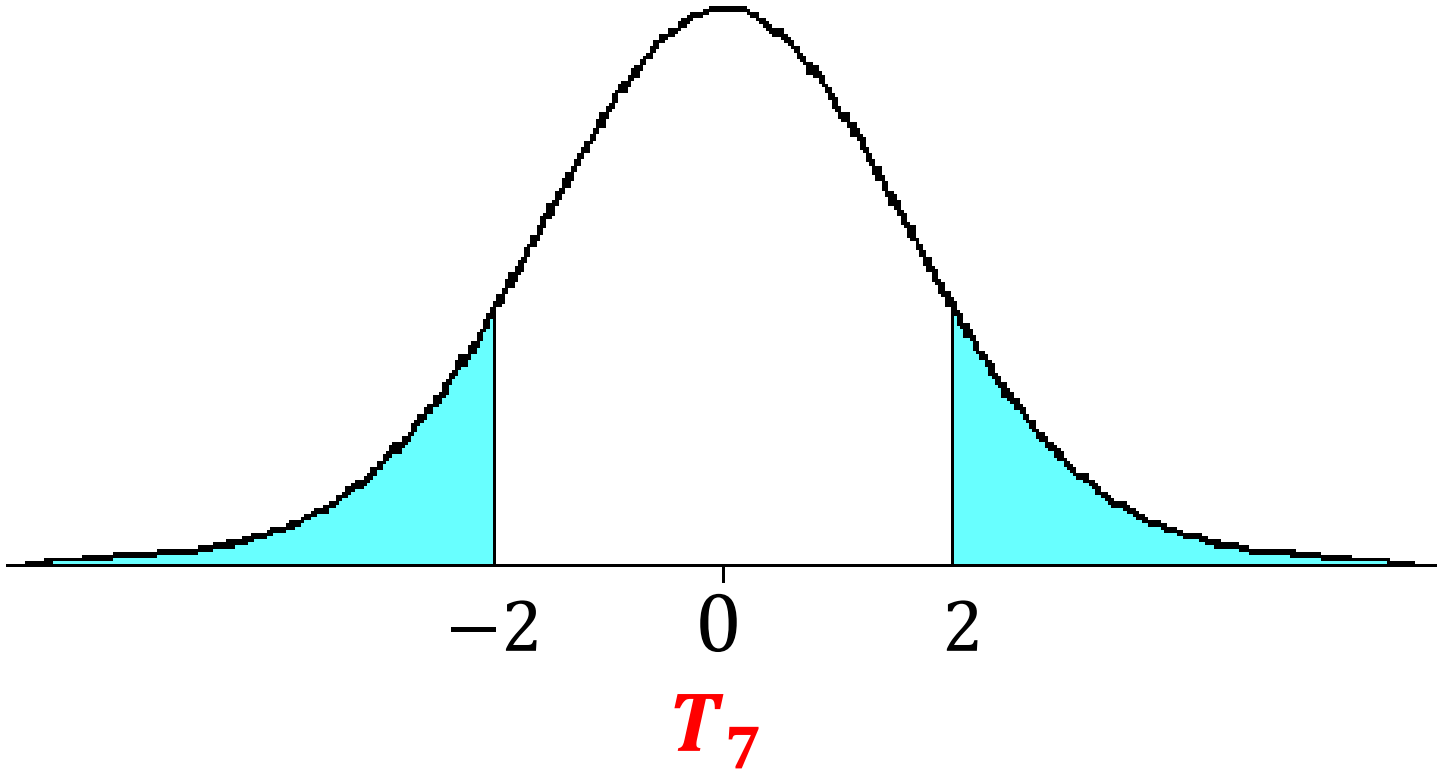
two-tailed test

Test statistic:

$$T = \frac{\overline{X} - \mu}{s/\sqrt{n}} = 2$$

Sample size:

$n = 8$



CASE WESTERN RESERVE UNIVERSITY

Look for two values in row 7. One value should be slightly less than 2, the other value should be slightly greater than 2

T-Table



	0.20	0.10	0.05	0.025	0.01	0.005	0.001	0.0005	0.0001
1	1.376	3.078	6.314	12.706	31.821	63.657	318.309	636.619	3183.099
2	1.061	1.886	2.920	4.303	6.965	9.925	22.327	31.599	70.700
3	0.978	1.638	2.353	3.182	4.541	5.841	10.215	12.924	22.204
4	0.941	1.533	2.132	2.776	3.747	4.604	7.173	8.610	13.034
5	0.920	1.476	2.015	2.571	3.365	4.032	5.893	6.869	9.678
6	0.906	1.440	1.943	2.447	3.143	3.707	5.208	5.959	8.025
7	0.896	1.415	1.895	2.365	2.998	3.499	4.785	5.408	7.063
8	0.889	1.397	1.860	2.306	2.896	3.355	4.501	5.041	6.442
9	0.883	1.383	1.833	2.262	2.821	3.250	4.297	4.781	6.010
10	0.879	1.372	1.812	2.228	2.764	3.169	4.144	4.587	5.694
11	0.876	1.363	1.796	2.201	2.718	3.106	4.025	4.437	5.453
12	0.873	1.356	1.782	2.179	2.681	3.055	3.930	4.318	5.263
13	0.870	1.350	1.771	2.160	2.650	3.012	3.852	4.221	5.111
14	0.868	1.345	1.761	2.145	2.624	2.977	3.787	4.140	4.985
15	0.866	1.341	1.753	2.131	2.602	2.947	3.733	4.073	4.880
16	0.865	1.337	1.746	2.120	2.583	2.921	3.686	4.015	4.791
17	0.863	1.333	1.740	2.110	2.567	2.898	3.646	3.965	4.714
18	0.862	1.330	1.734	2.101	2.552	2.878	3.610	3.922	4.648
19	0.861	1.328	1.729	2.093	2.539	2.861	3.579	3.883	4.590
20	0.860	1.325	1.725	2.086	2.528	2.845	3.552	3.850	4.539

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$H_0: \mu = 3.72$

$H_a: \mu \neq 3.72$

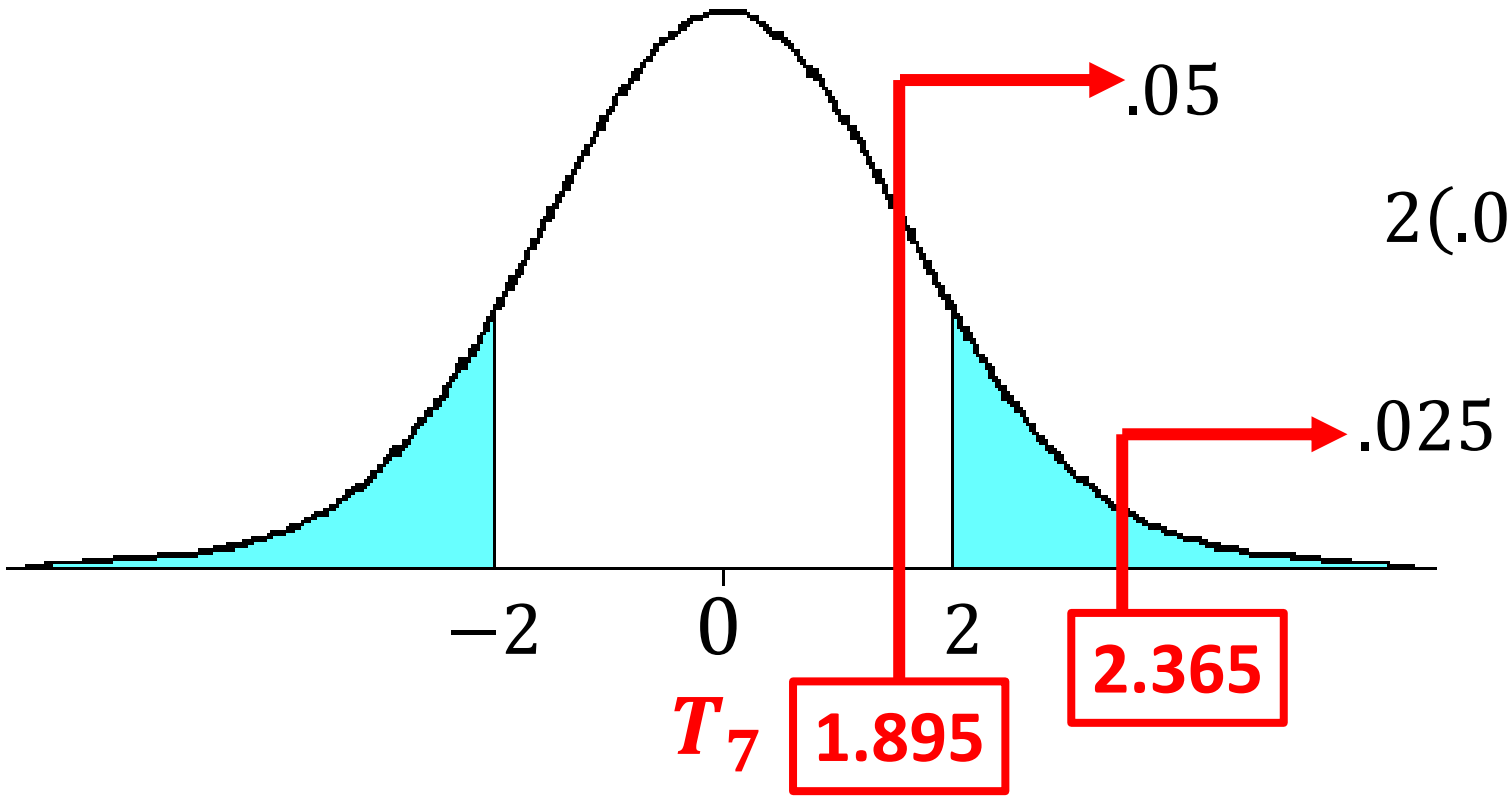
two-tailed test

Test statistic:

$$T = \frac{\overline{X} - \mu}{s/\sqrt{n}} = 2$$

Sample size:

$$n = 8$$



$$.025 < \frac{\text{p-value}}{2} < .05$$

$$2(.025) < 2\left(\frac{\text{p-value}}{2}\right) < 2(.05)$$

$$.05 < \text{p-value} < .10$$

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$H_0: \mu = 3.72$

$H_a: \mu \neq 3.72$

two-tailed test

Test statistic:

$$T = \frac{\overline{X} - \mu}{s/\sqrt{n}} = 2$$

Sample size:

$n = 8$

$.05 < \text{p-value} < .10$

Was H_0 rejected?

We don't know. Knowing the value of α might tell us.

If $\alpha = .04$ then H_0 was not rejected because p-value $> \alpha$

If $\alpha = .11$ then H_0 was rejected because p-value $< \alpha$

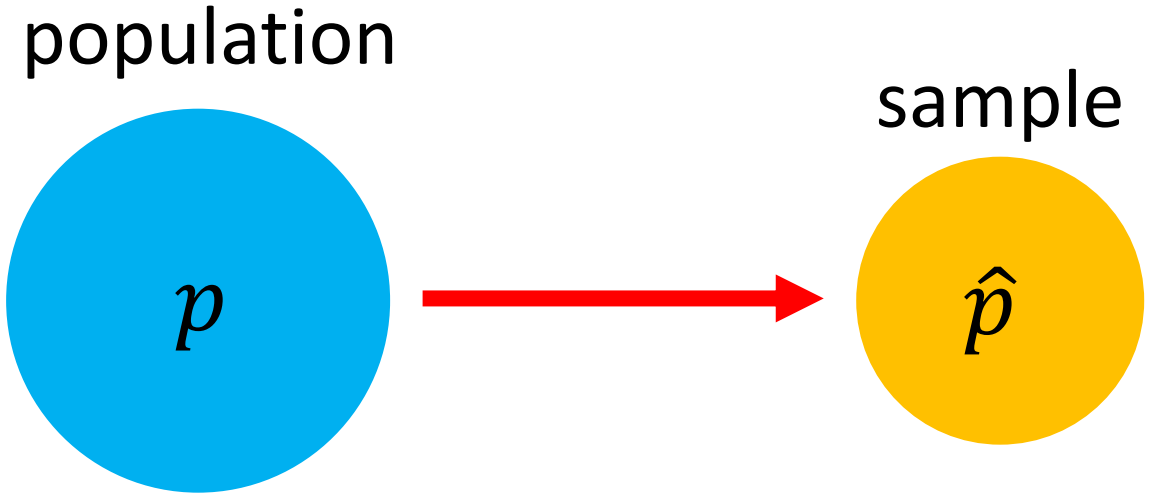
If $\alpha = .08$ then we still don't know if H_0 was rejected.

ONE SAMPLE Z TEST FOR P

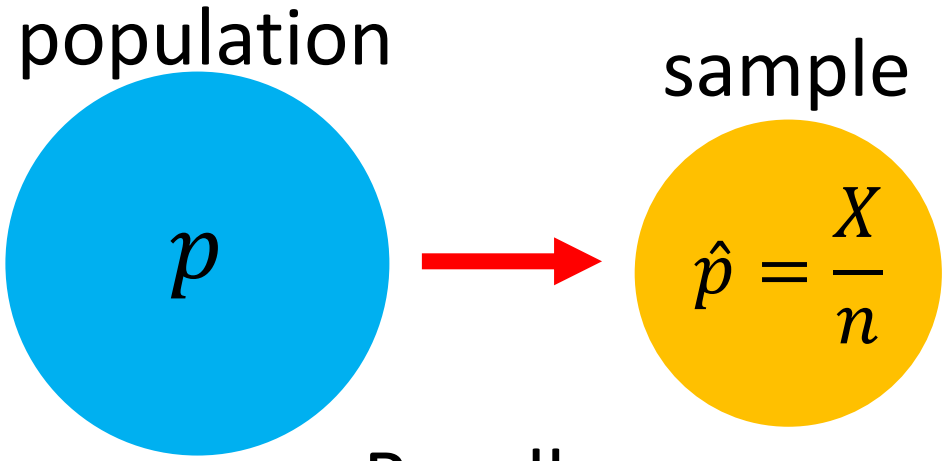
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think beyond the possible℠

Recall:

A proportion is simply a percentage expressed as a fraction or decimal between zero and one.



The sample proportion $\hat{p}$ is an estimator of the unknown population proportion p.



$$E(\hat{p}) = p$$

$$SD(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

Recall:

$$X \sim BIN(n, p)$$

X counts the # of successes in n trials.

p = P(success). This is the same p.

$$\mu = E(X) = np$$

$$\sigma^2 = Var(X) = np(1-p)$$

A $(1-\alpha)100\%$ confidence interval for p $(n \geq 30)$

$$\hat{p} \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

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one sample Z test for p

$H_0: p = p_0$

$H_a: p < p_0$ or $H_a: p > p_0$ or $H_a: p \neq p_0$

left-tailed | right-tailed | two-tailed

All Z and T test statistics follow a basic recipe:

$$\frac{\text{point estimate} - \text{hypothesized parameter value}}{SD(\text{estimator})}$$

one sample Z test for p

$$H_0: p = p_0$$

All Z and T test statistics follow a basic recipe:

$$\frac{\text{point estimate } - \text{hypothesized parameter value}}{SD(\text{estimator})}$$

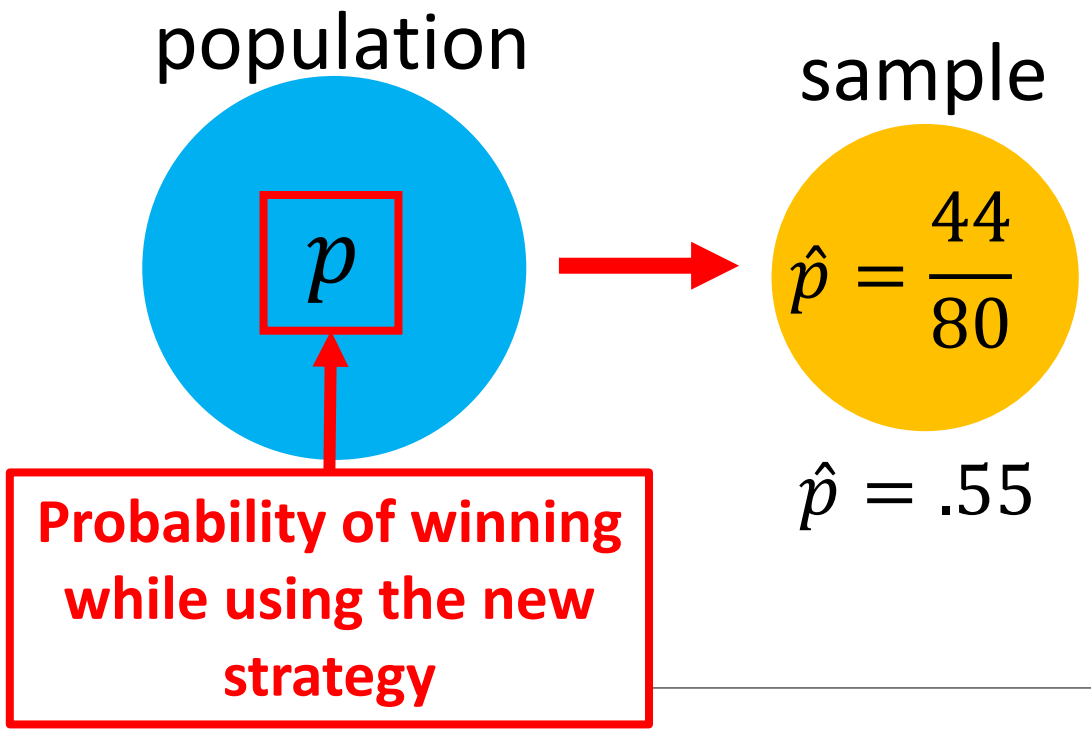
$$E(\hat{p}) = p$$

$$SD(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Historically, a certain gambling game has $p = P(\text{win}) = .49$

Your friend has developed a new strategy that he hopes (suspects) will increase the probability of winning. To test the new strategy, he plays 80 games, and wins 44 of them.



$H_0: p = .49$

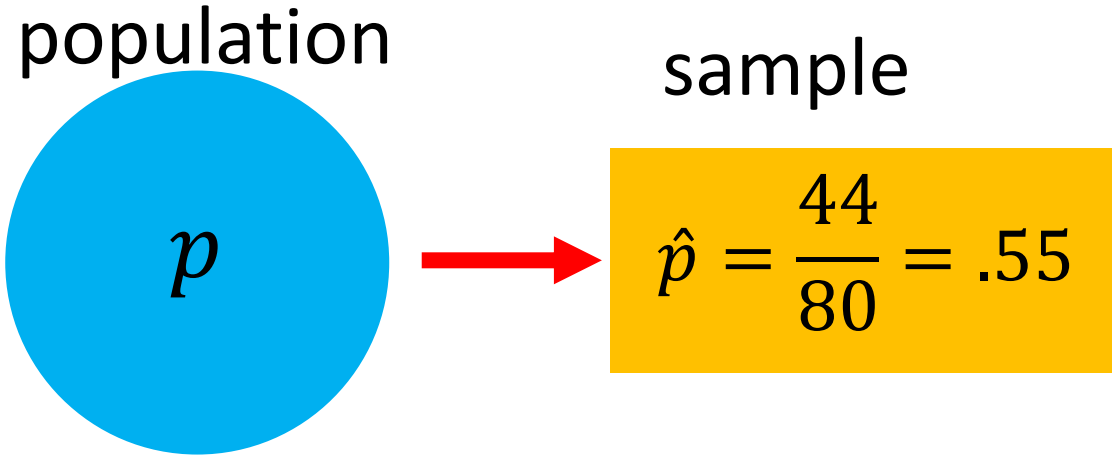
(the new strategy does not improve the chance of winning)

$H_a: p > .49$

(the new strategy improves the chance of winning)

$\alpha = .05$

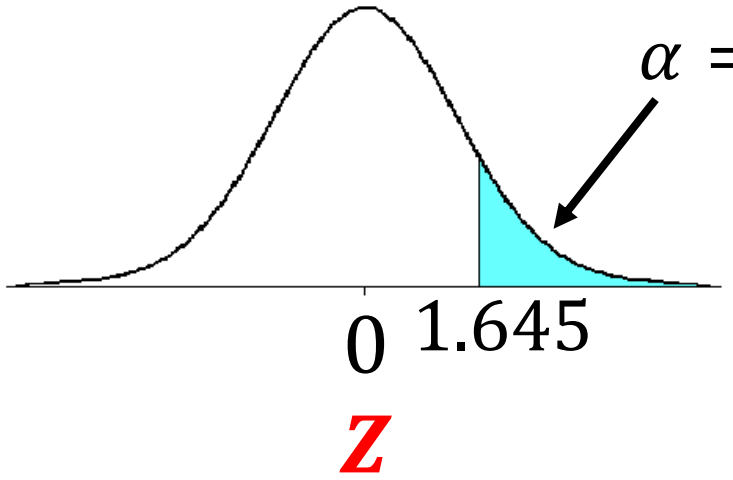
CASE WESTERN RESERVE UNIVERSITY



$H_0: p = .49$
(the new strategy does not improve the chance of winning)

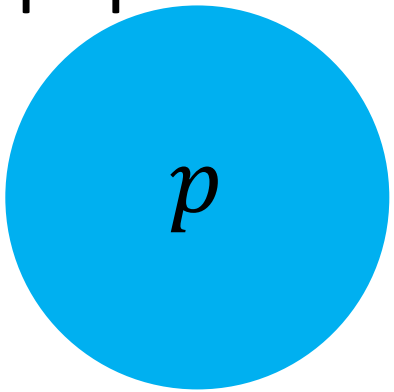
$H_a: p > .49$
(the new strategy improves the chance of winning)

Rejection Region: $Z > 1.645$



$\alpha = .05$

population

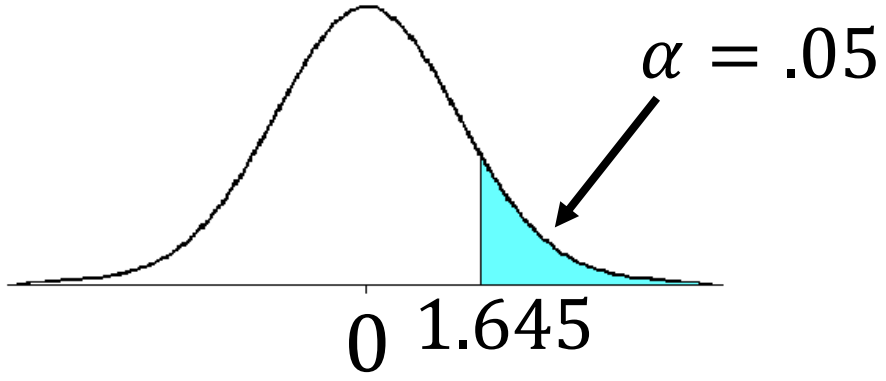


sample

$$\hat{p} = \frac{44}{80} = .55$$

$$H_0: p = .49$$
$$H_a: p > .49$$
$$\alpha = .05$$

Rejection Region: $Z > 1.645$



Test Statistic: $Z = \dfrac{\hat{p} - p_0}{\sqrt{\dfrac{p_0(1-p_0)}{n}}} = \dfrac{.55 - .49}{\sqrt{\dfrac{.49(.51)}{80}}} \approx 1.07$

$H_0: p = .49$

$H_a: p > .49$

$\alpha = .05$

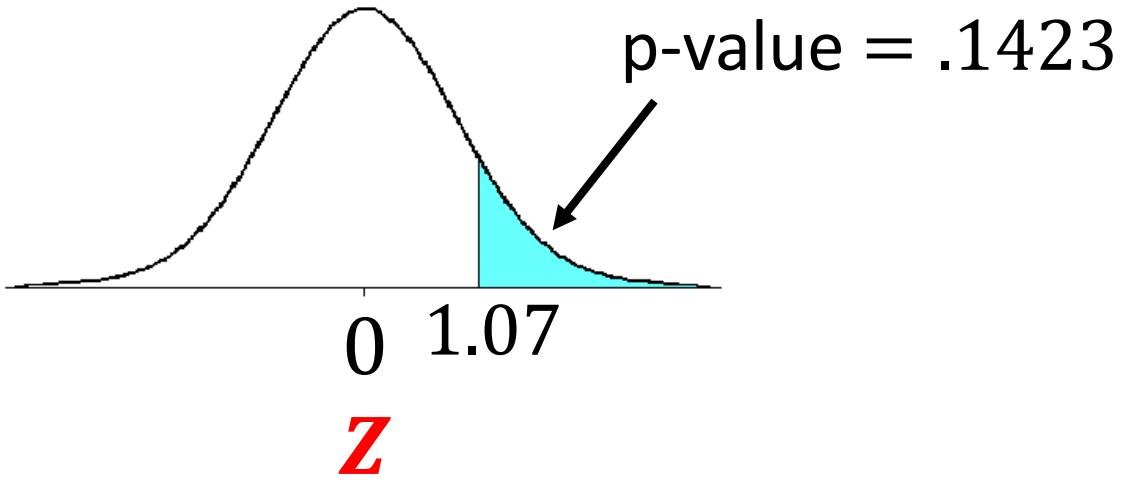
Rejection Region:

$Z > 1.645$

Test Statistic: $Z = \dfrac{\hat{p} - p_0}{\sqrt{\dfrac{p_0(1-p_0)}{n}}} = \dfrac{.55 - .49}{\sqrt{\dfrac{.49(.51)}{80}}} \approx 1.07$

Don't reject H_0

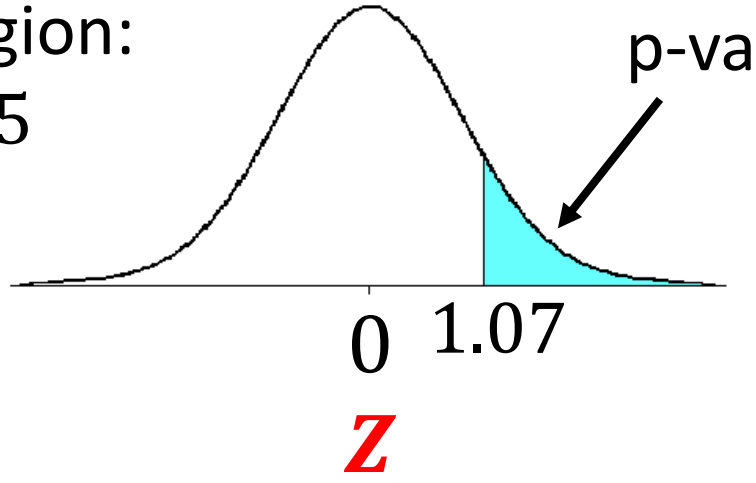
Since we didn't reject H_0, we know p-value $> \alpha$



$$H_0: p = .49$$
$$H_a: p > .49$$
$$\alpha = .05$$

Rejection Region:

$$Z > 1.645$$



$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{.55 - .49}{\sqrt{\frac{.49(.51)}{80}}} \approx 1.07$$

Don't reject H_0

"If a large group of gamblers do not use your new strategy (and win with a probability of .49), about 14.23% of them will do just as good as you did (or even better) just out of luck."

wins	n	$\hat{p}$	Z	p-value
44	80	.55	1.07	.1423
55	100	.55	1.20	.1151
110	200	.55	1.70	.0446
275	500	.55	2.68	.0037
550	1000	.55	3.80	$\approx \frac{1}{14000}$

At what p-value do you become convinced $p > .49$?

That should be your choice for α

THE BETA AND POWER FUNCTIONS- OPTIONAL TOPIC

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Recall:

β = P(Type 2 error) = P(do not reject a false H_0)

$1 - \beta$ = Power = P(correctly reject a false H_0)

In this topic we will talk about two functions:

$\beta(x)$ = P(not rejecting H_0 when the unknown parameter equals a specific value named x)

Power(x) = P(reject H_0 when the unknown parameter equals x)

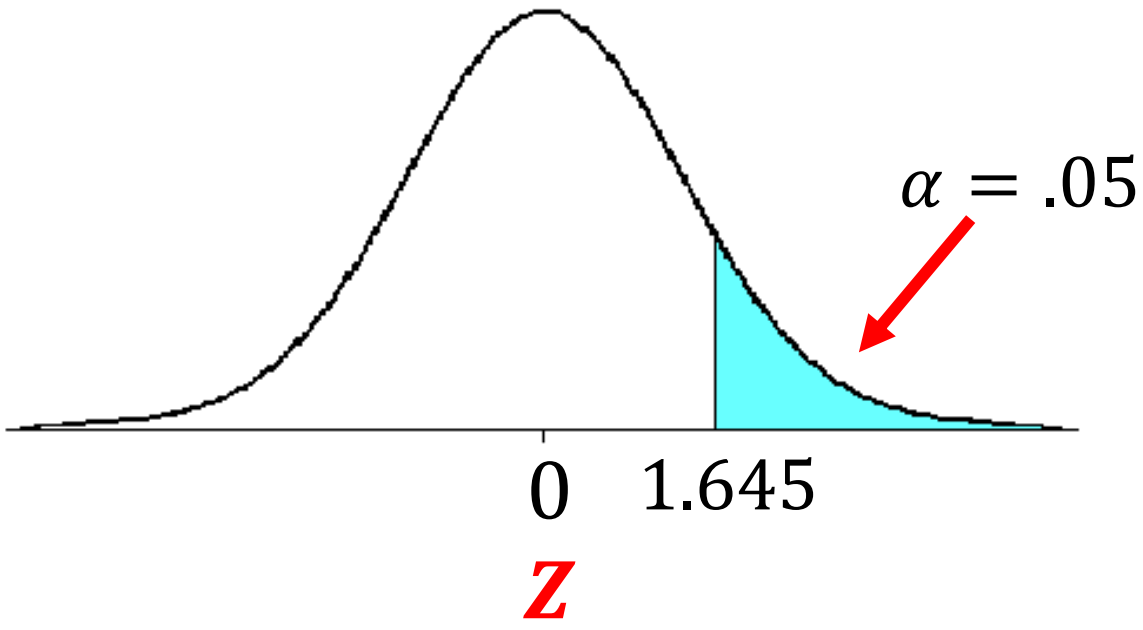
Power(x)= $1 - \beta(x)$

Suppose someone is conducting a one-sample Z-test for μ (σ known to be 4)

$H_0: \mu = 100$ cm

$H_a: \mu > 100$ cm

$\alpha = .05$



Suppose someone is conducting a one-sample Z-test for μ (σ known to be 4)

$$H_0: \mu = 100 \text{ cm}$$

$$H_a: \mu > 100 \text{ cm}$$

$$\alpha = .05$$

Rejection Region:

$$Z_{TS} = \frac{\overline{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{\overline{X} - 100}{\frac{4}{\sqrt{n}}} > 1.645$$

$$H_0: \mu = 100 \text{ cm}$$
$$H_a: \mu > 100 \text{ cm}$$
$$\alpha = .05$$

Rejection Region:

$$Z_{TS} = \frac{\overline{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{\overline{X} - 100}{\frac{4}{\sqrt{n}}} > 1.645$$

If $n = 10$ and $\mu_a = 101$, what is β = P(Type 2 error)?

hypothesized value: $\mu_0 = 100$

actual value: $\mu_a = 101$

$$H_0: \mu = 100 \text{ cm}$$

$$H_a: \mu > 100 \text{ cm}$$

$$\alpha = .05$$

$$Z_{TS} = \frac{\overline{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{\bar{X} - 100}{\frac{4}{\sqrt{10}}} > 1.645$$

If $n = 10$ and $\mu_a = 101$, what is β = P(Type 2 error)?

hypothesized value: $\mu_0 = 100$

actual value: $\mu_a = 101$

$$P(Z_{TS} < 1.645) = P\left(\frac{\overline{X} - 100}{\frac{4}{\sqrt{10}}} < 1.645\right) = P\left(\overline{X} - 100 < 1.645\frac{4}{\sqrt{10}}\right)$$

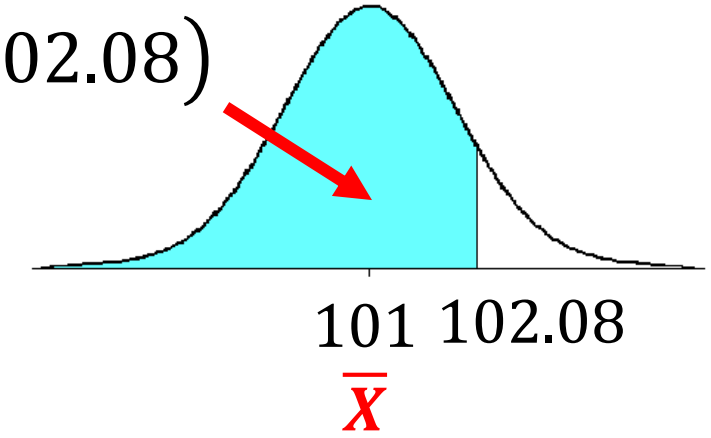
$$= P\left(\overline{X} < 100 + 1.645\frac{4}{\sqrt{10}}\right) = P\left(\overline{X} < 102.08\right)$$

If $n = 10$ and $\mu_a = 101$, what is β = P(Type 2 error)?

hypothesized value: $\mu_0 = 100$

actual value: $\mu_a = 101$

$P(\overline{X} < 102.08)$



$$\overline{X} \sim N\left(\mu_a = 101, \frac{\sigma}{\sqrt{n}} = \frac{4}{\sqrt{10}}\right)$$

$$P(\overline{X} < 102.08) = P\left(Z < \frac{102.08 - 101}{\frac{4}{\sqrt{10}}}\right) = P(Z < 0.85) = .8023$$

There is a good chance the 1 cm difference between $\mu_0 = 100$ and $\mu_a = 101$ will <u>not</u> be detected.

$H_0: \mu = 100$ cm
$H_a: \mu > 100$ cm

$\alpha = .05$

σ known to be 4

If $n = 10$ and $\mu_a = 101$, β = P(Type 2 error) = $\beta(101) = .8023$

What if $n = 15$ and $\mu_a = 101$, $\beta(101) = ?$

$$P(Z_{TS} < 1.645) = P\left(\frac{\overline{X} - 100}{\frac{4}{\sqrt{15}}} < 1.645\right) = P\left(\overline{X} - 100 < 1.645\frac{4}{\sqrt{15}}\right)$$

$$= P\left(\overline{X} < 100 + 1.645\frac{4}{\sqrt{15}}\right) = P\left(\overline{X} < 101.70\right)$$

$H_0: \mu = 100$ cm
$H_a: \mu > 100$ cm

$\alpha = .05$

σ known to be 4

If $n = 10$ and $\mu_a = 101$, β = P(Type 2 error) = $\beta(101) = .8023$

What if $n = 15$ and $\mu_a = 101$, β = ?

$$P(\overline{X} < 101.70) = P\left(Z < \frac{101.70 - 101}{\frac{4}{\sqrt{15}}}\right) = P(Z < 0.68) = .7517$$

$H_0: \mu = 100$ cm
$H_a: \mu > 100$ cm

$\alpha = .05$

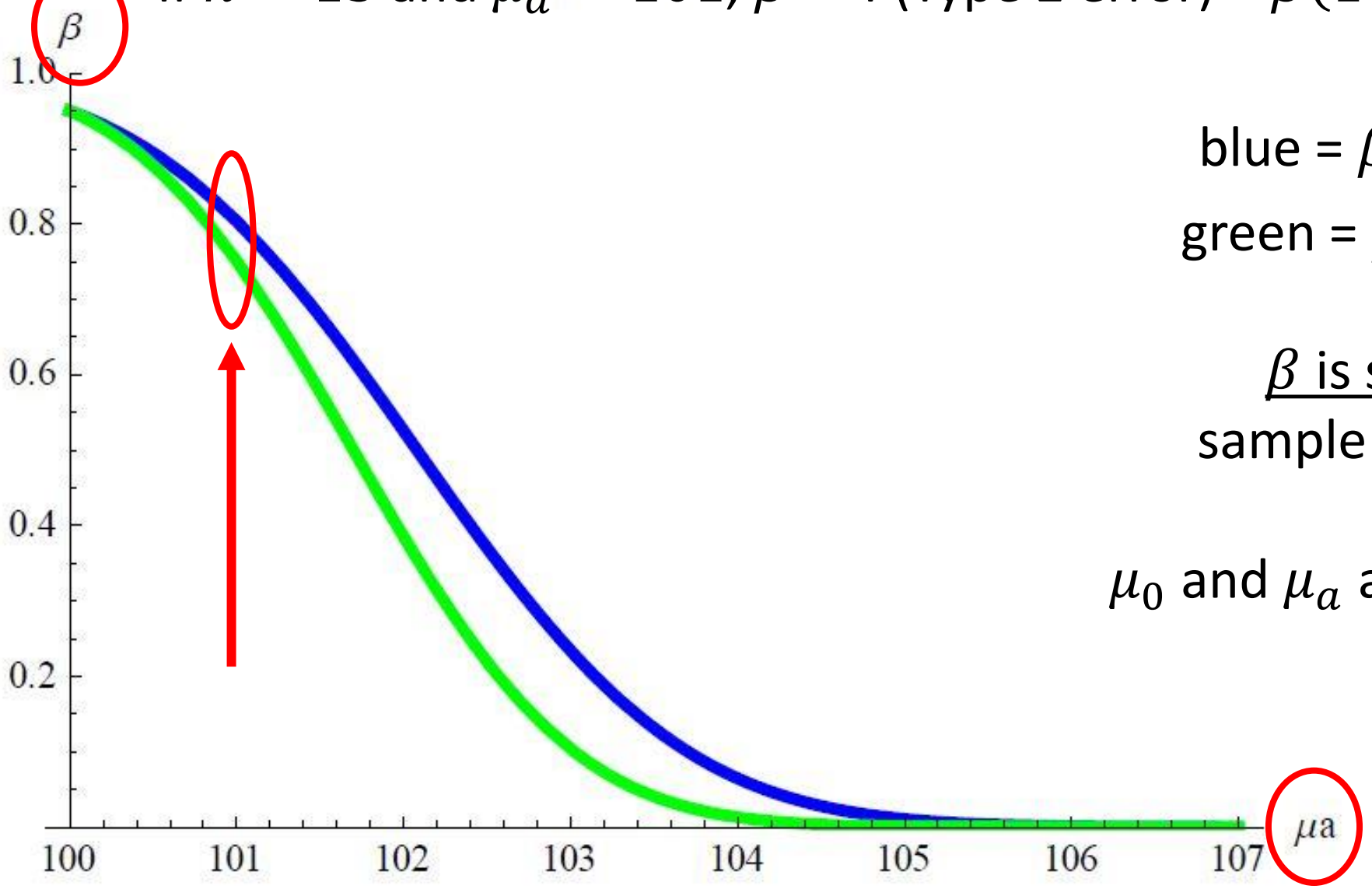
σ known to be 4

If $n = 10$ and $\mu_a = 101$, β = P(Type 2 error) = $\beta(101) = .8023$

If $n = 15$ and $\mu_a = 101$, β = P(Type 2 error) = $\beta(101) = .7517$

If $n = 10$ and $\mu_a = 101$, β = P(Type 2 error) = $\beta(101) = .8023$

If $n = 15$ and $\mu_a = 101$, β = P(Type 2 error) = $\beta(101) = .7517$



blue = β when $n = 10$

green = β when $n = 15$

<u>β is smaller if</u>:

sample size is larger

or

μ_0 and μ_a are further apart

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In this topic we will talk about two functions:

$\beta(x) =$ P(not rejecting H_0 when the unknown parameter equals a specific value named x)

Power(x) = P(reject H_0 when the unknown parameter equals x)

Power(x)$= 1 - \beta(x)$

Power(x) = P(reject H_0 when the unknown parameter equals x)

A perfect hypothesis test would have:

P(correctly reject a false H_0) = 1

P(incorrectly reject a true H_0) = 0

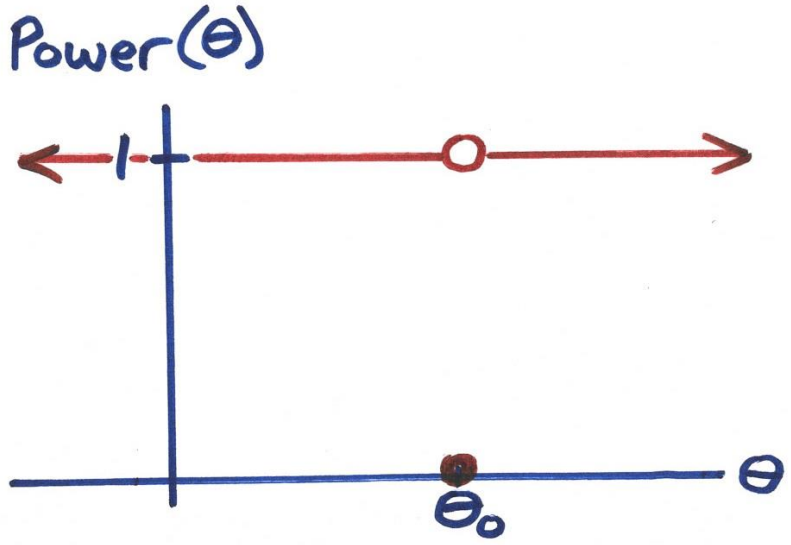
In general, a two-sided hypothesis test for an unknown parameter can be written as:

H_0: $\theta = \theta_0$

H_a: $\theta \neq \theta_0$

$$P(\text{reject } H_0 \mid H_0 \text{ is true}) = 0$$
$$P(\text{reject } H_0 \mid H_0 \text{ is false}) = 1$$

Although a perfect hypothesis test does not exist in practice, looking at this graph can help us evaluate the quality of other hypothesis tests and their power functions.

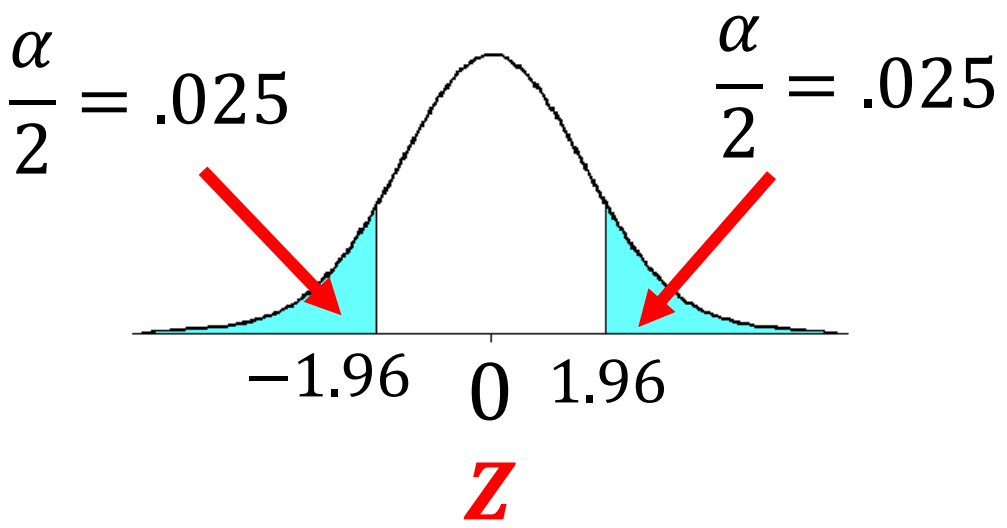


A one-sample Z-test for μ

μ unknown, σ known to be 1

$$H_0: \mu = 7$$
$$H_a: \mu \neq 7$$
$$\sigma = 1$$
$$\alpha = .05$$

RR: $Z > 1.96$ or $Z < -1.96$



If $n = 1$ and $\mu_a = 8$

Find Power(8) = the probability of correctly rejecting H_0 if the actual value of μ is 8

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$$H_0: \mu = 7$$
$$H_a: \mu \neq 7$$
$$\sigma = 1$$

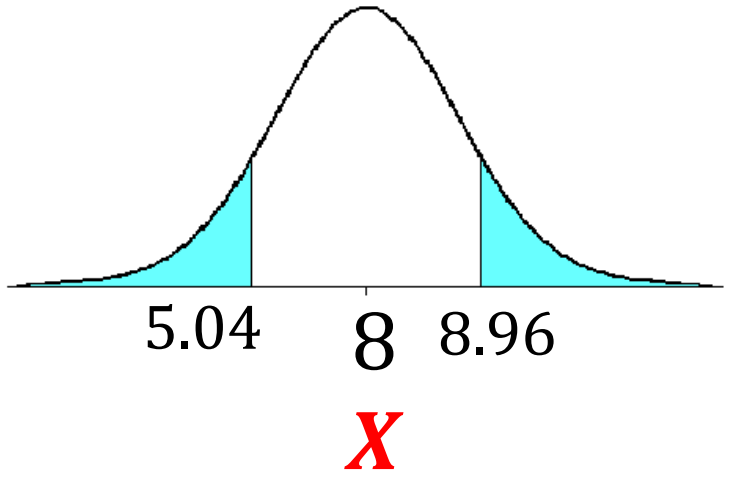
$$\alpha = .05$$
RR: $Z > 1.96$ or $Z < -1.96$

If $n = 1$ and $\mu_a = 8$

Power(8) = the probability of correctly rejecting H_0 if the actual value of μ is 8

$$\text{Power}(8) = P\left(\frac{X-7}{\frac{1}{\sqrt{1}}} > 1.96 \text{ or } \frac{X-7}{\frac{1}{\sqrt{1}}} < -1.96\right) = P(X > 8.96 \text{ or } X < 5.04)$$

$$X \sim N(\mu = 8, \sigma = 1) \rightarrow$$

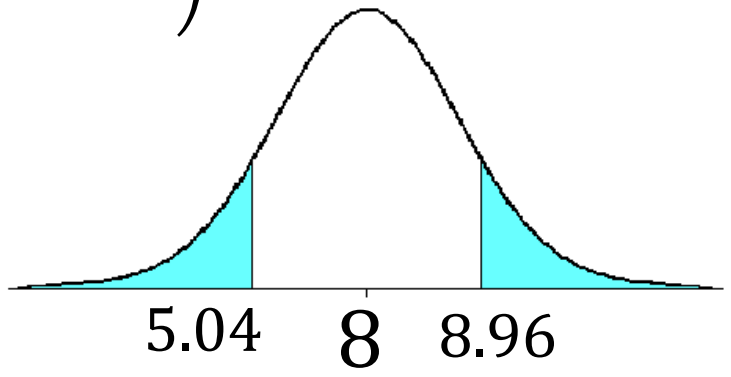


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Power(8) = the probability of correctly rejecting H_0 if the actual value of μ is 8

$$\text{Power}(8) = P\left(\frac{X-7}{\frac{1}{\sqrt{1}}} > 1.96 \text{ or } \frac{X-7}{\frac{1}{\sqrt{1}}} < -1.96\right) = P(X > 8.96 \text{ or } X < 5.04)$$

$$X \sim N(\mu = 8, \sigma = 1) \rightarrow$$



$$P(X > 8.96 \text{ or } X < 5.04) = P\left(Z > \frac{8.96-8}{\frac{1}{\sqrt{1}}} \text{ or } Z < \frac{5.04-8}{\frac{1}{\sqrt{1}}}\right)$$

$$= P(Z > 0.96 \text{ or } Z < -2.96) = .1685 + .0015 = .1700$$

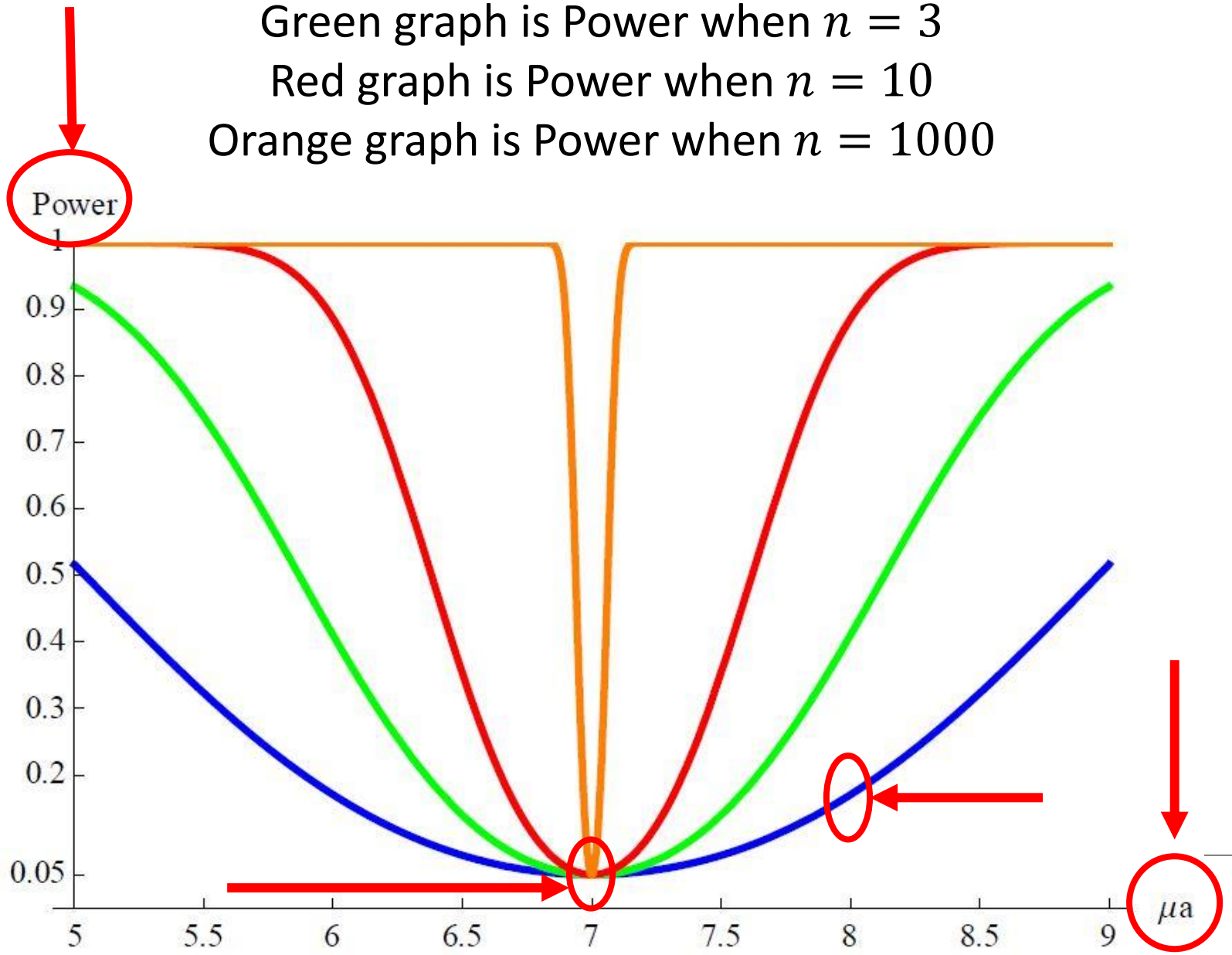
Power function at different choices of n

Blue graph is Power when $n = 1$

Green graph is Power when $n = 3$

Red graph is Power when $n = 10$

Orange graph is Power when $n = 1000$



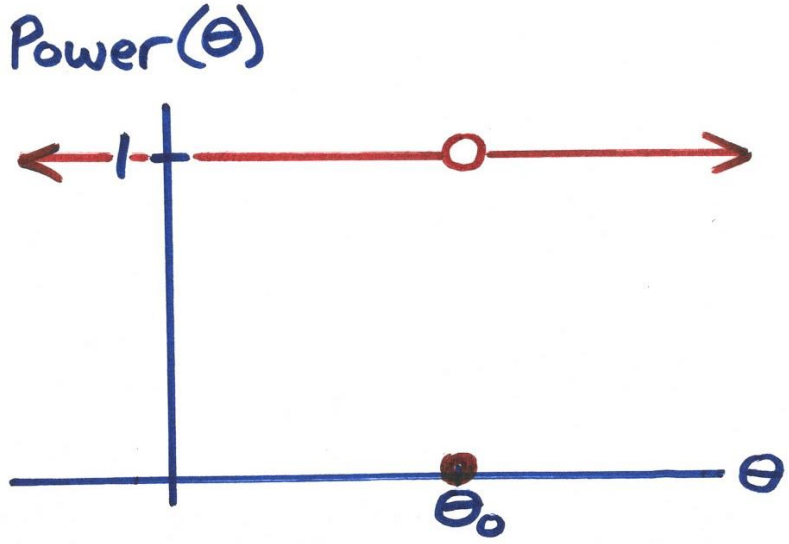
$$Y \sim N(\mu\ unknown, \sigma = 1)$$

$$H_0: \mu = 7$$

$$H_a: \mu \neq 7$$

$$\alpha = .05$$

$$RR: Z > 1.96\ or\ Z < -1.96$$



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DEFINITION OF T

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Let Z be a standard normal random variable.

$$Z \sim N(\mu = 0, \sigma = 1)$$

Let W be a χ^2 random variable with v degrees of freedom.

If Z and W are independent, then

$$T = \frac{Z}{\sqrt{W/v}} \sim t_v$$

If Z and W are independent, then

$$T = \frac{Z}{\sqrt{W/v}} \sim t_v$$

Suppose you have a random sample from a normal distribution.

$$X \sim N(\mu, \sigma)$$

$$\frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim Z \qquad \frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$$

$$T = \frac{Z}{\sqrt{W/v}} \sim t_v \qquad \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim Z \qquad \frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$$

$$T = \frac{\frac{\overline{X} - \mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{\frac{(n-1)s^2}{\sigma^2}}{n-1}}} = \left(\frac{\overline{X} - \mu}{\sigma}\right)\left(\frac{\sqrt{n}}{1}\right)\frac{\sqrt{\sigma^2}}{\sqrt{s^2}} = \frac{\overline{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

$$\frac{\overline{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$